

Statistical learning theory

Foundations of AI

Carlos Cotrini

UZH CAS Gamechanger AI

26 September 2026

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Section 1

Empirical risk minimisation

Vladimir Vapnik



Vladimir Vapnik



Second edition, 2006

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- **Fitting a price.** The score is the squared miss, the one this room has used all day.
- **Estimating a probability.** The score is how surprised the model is by what actually happened.
- Three problems that had lived in different books. The score says which one you are doing, and what happens to the score afterwards never changes.

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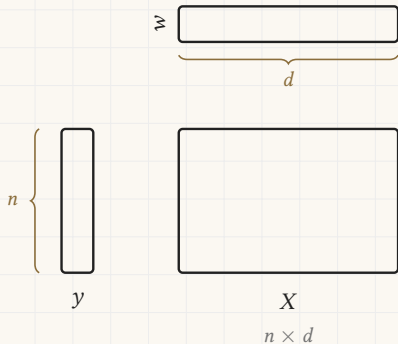
2

Section 2

The loop we have been running

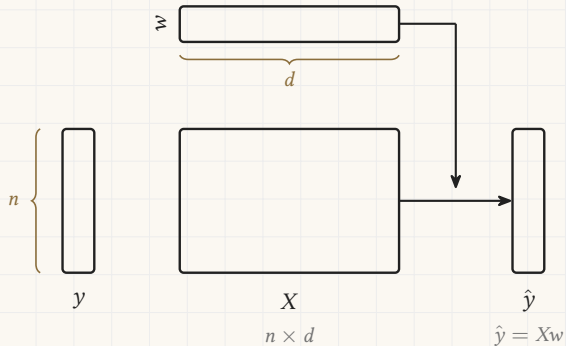
Linear regression with gradient descent

Two o'clock, in one picture. Each box is a piece of data, and its shape is the shape of the data.



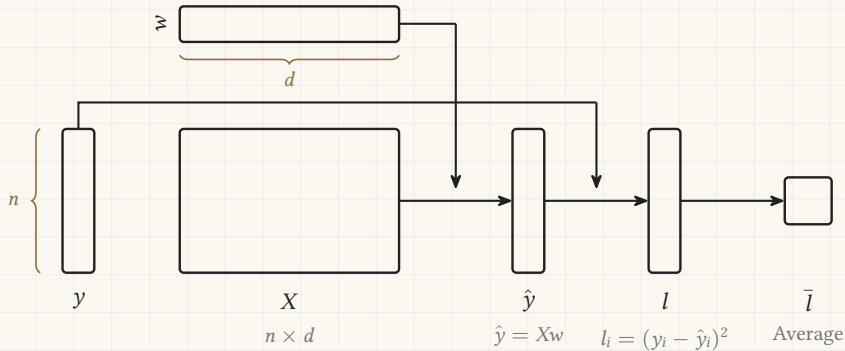
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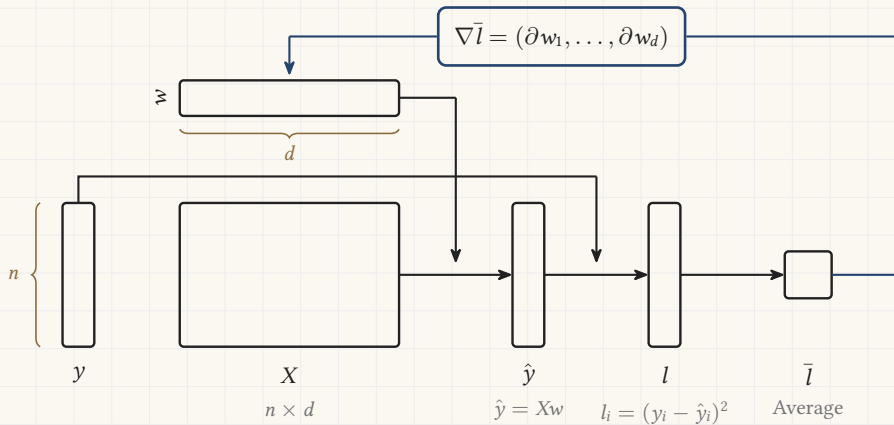
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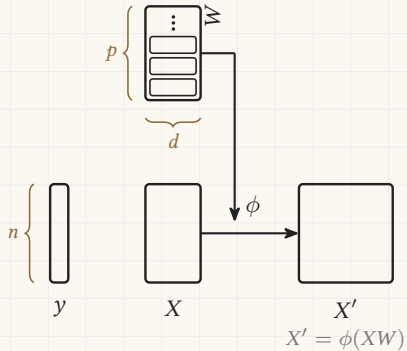


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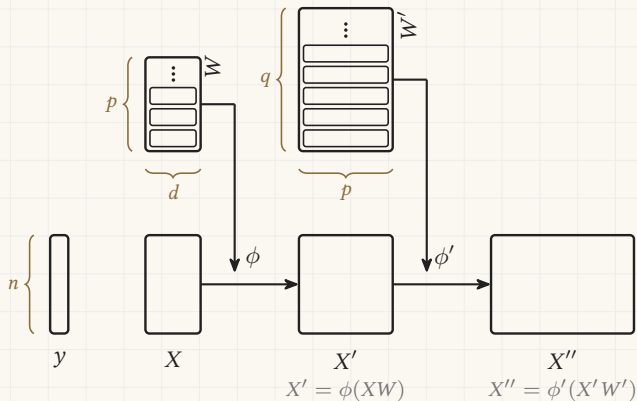
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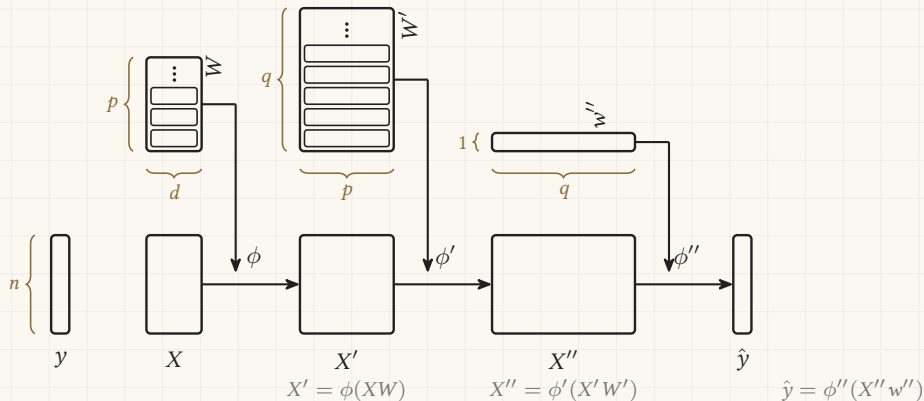
A neural network



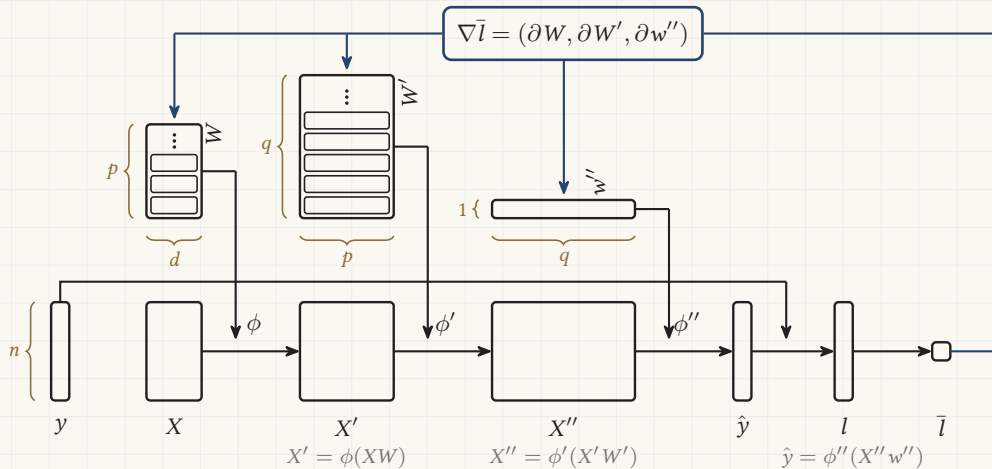
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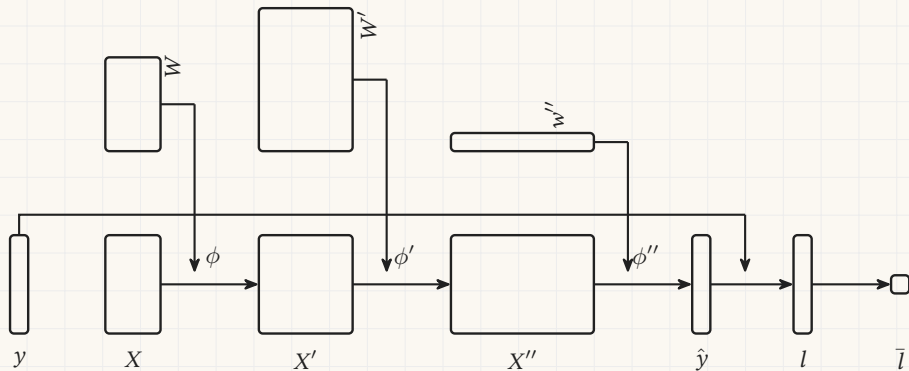


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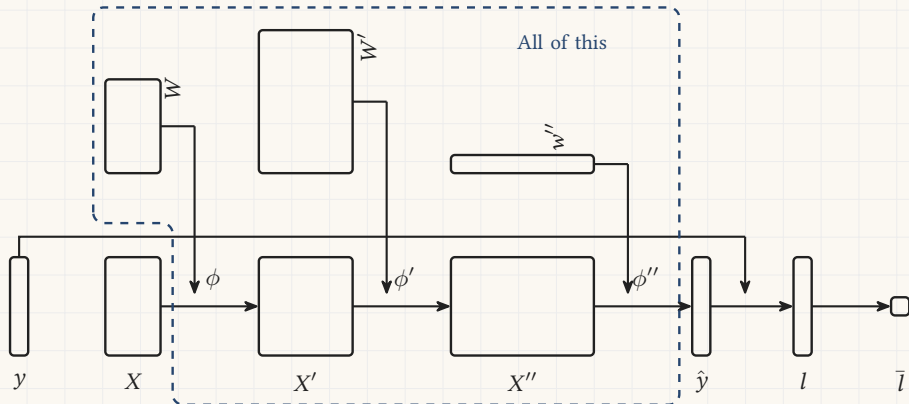
Everything between the data and the estimate

The three weight blocks and the two tables between them are the model. Give the whole of it one name.



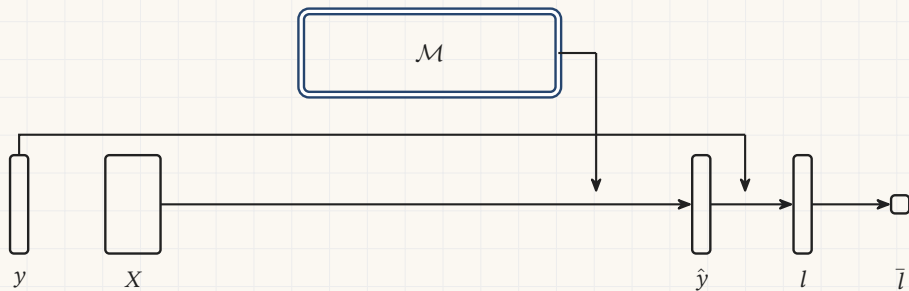
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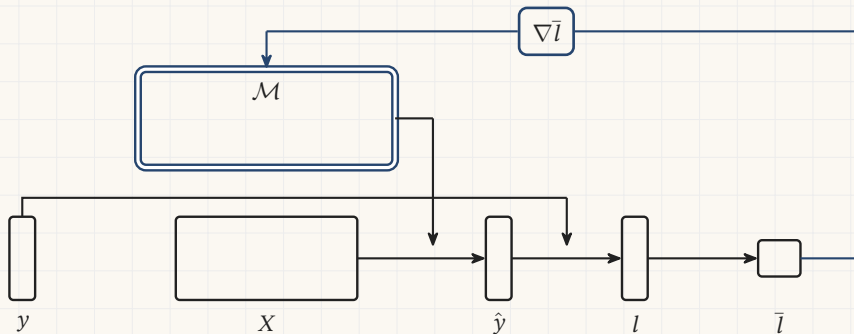
Section 3

The same loop, elsewhere

The loop, with the model as one box

The same picture, with $\mathcal{M}$ where the network was.

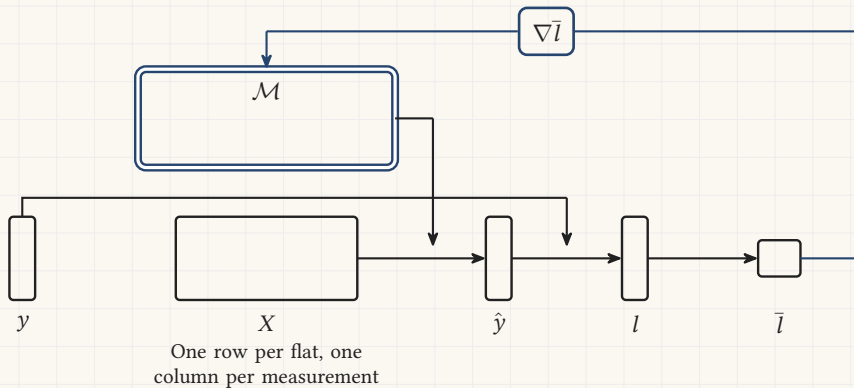
The five flats are the ones from this morning, and the network is the one from this afternoon.



The loop, with the model as one box

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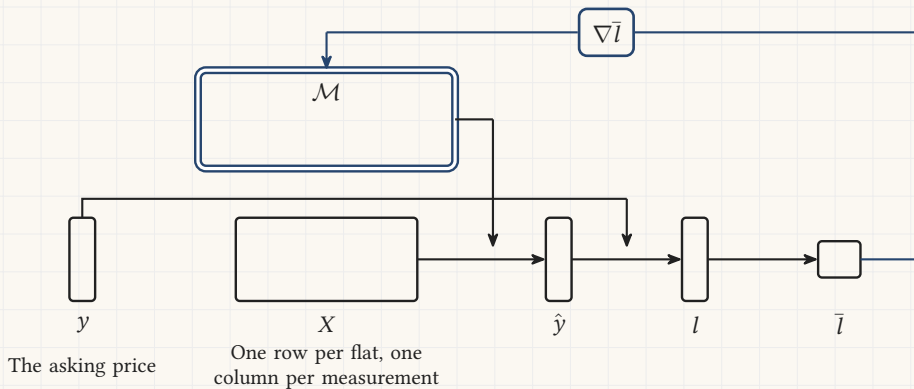
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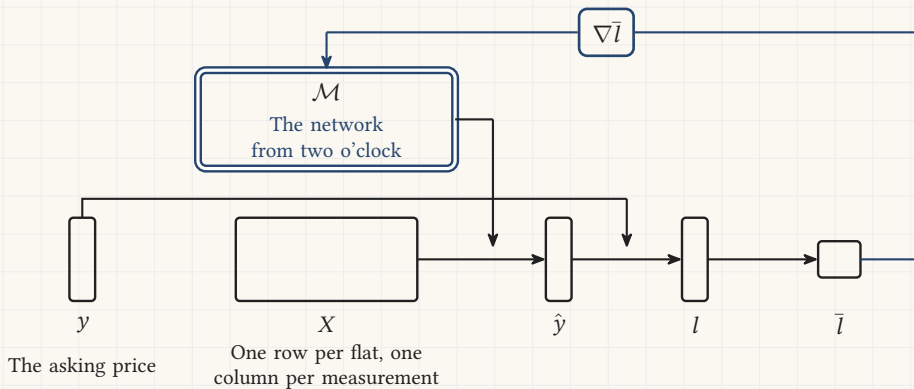
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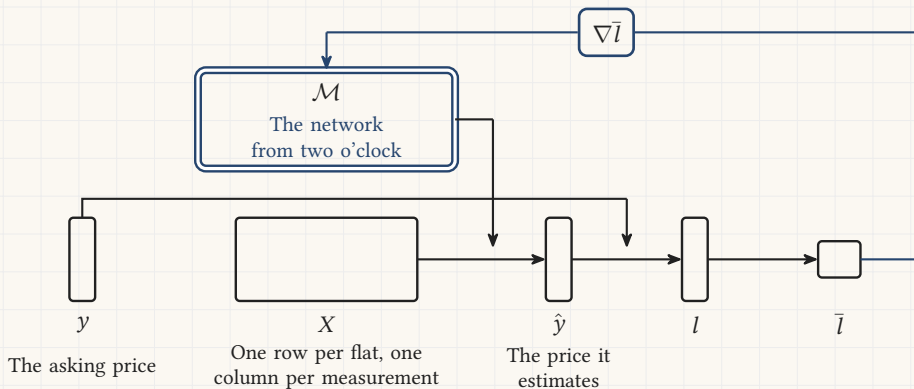
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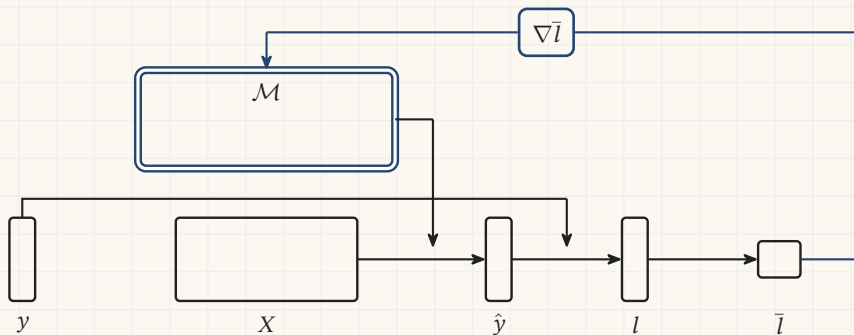
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Large language models

A row is text with its next token cut off. Every position of every document gives one.

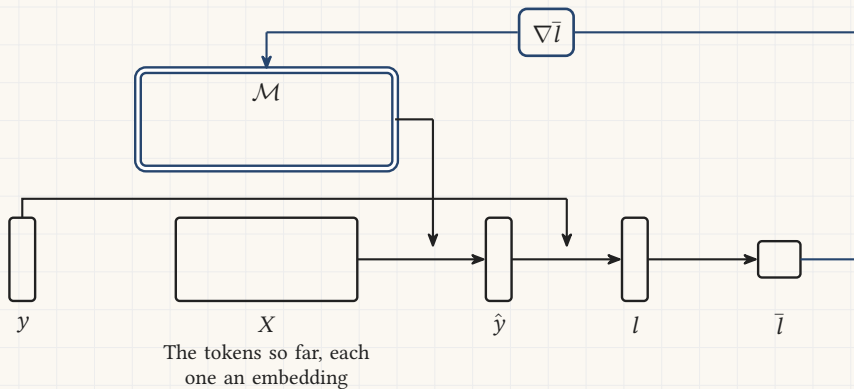
The box holds a transformer. Vaswani and others, *Attention is all you need*, 2017.



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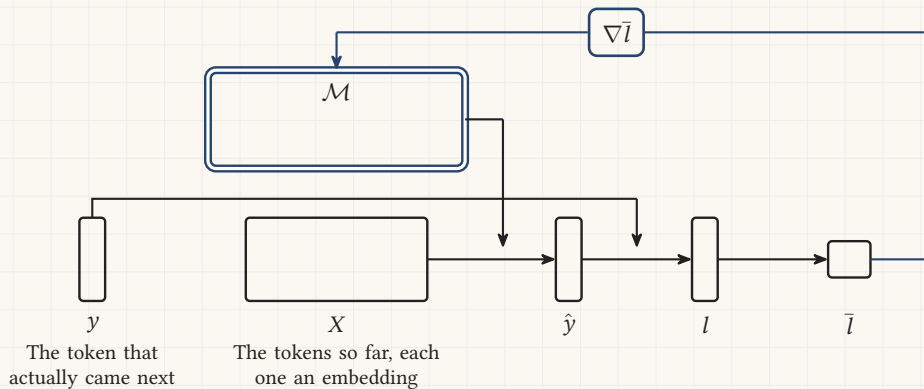
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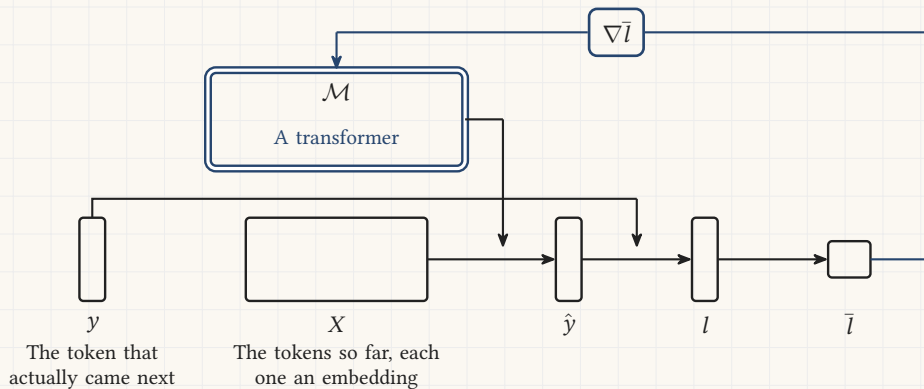
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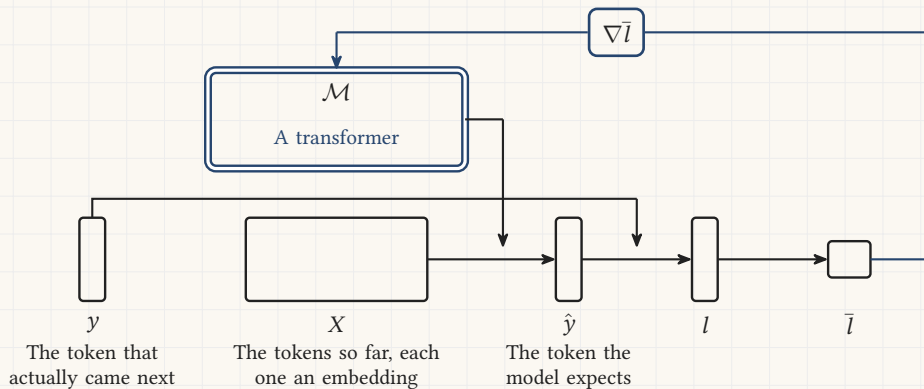


Image generation

Add a measured amount of noise to a picture, then ask which noise was added.

The loss here really is the squared miss. Ho, Jain and Abbeel, 2020; Rombach and others, 2022.

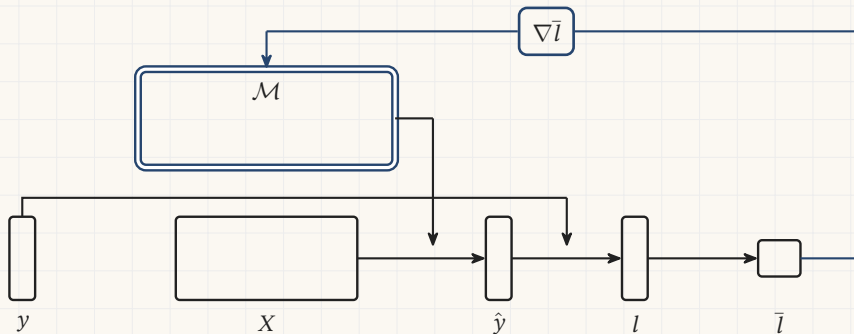


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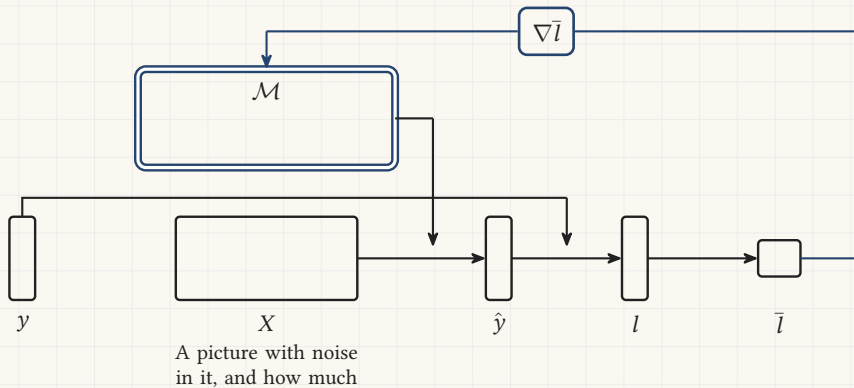


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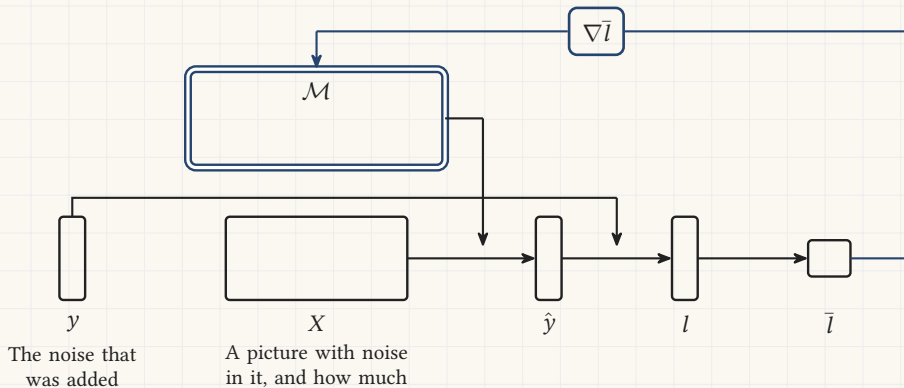


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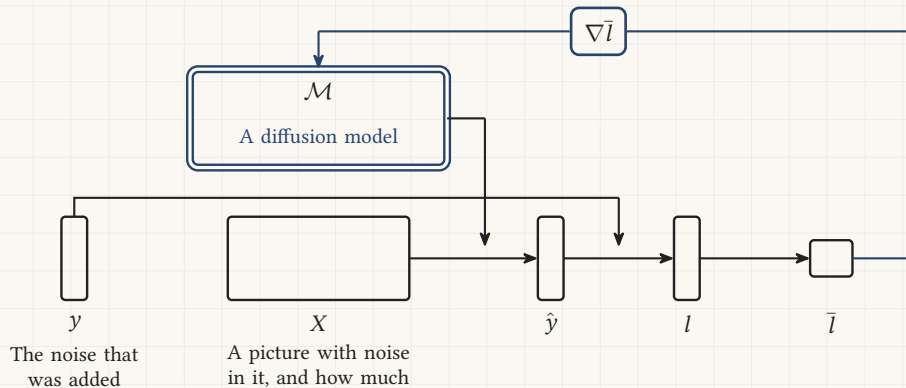
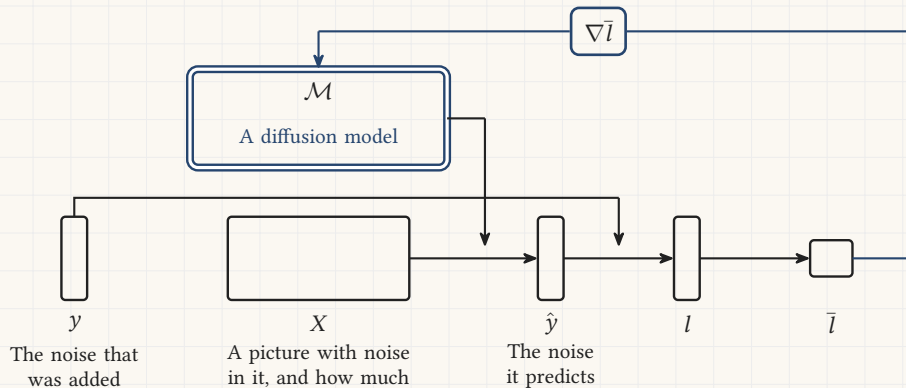


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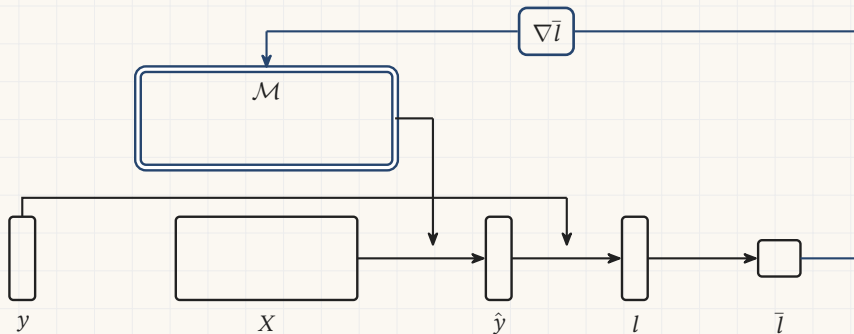
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Transcription

Every recording that already carries a transcript is a row. There are 680 000 hours.

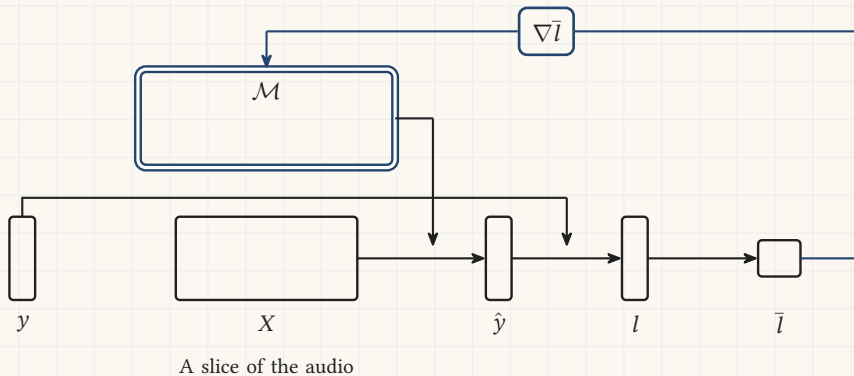
Whisper. Radford and others, ICML 2023.



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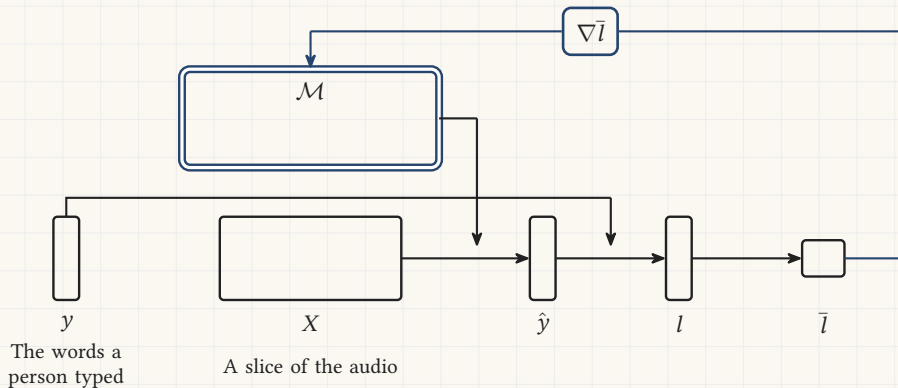
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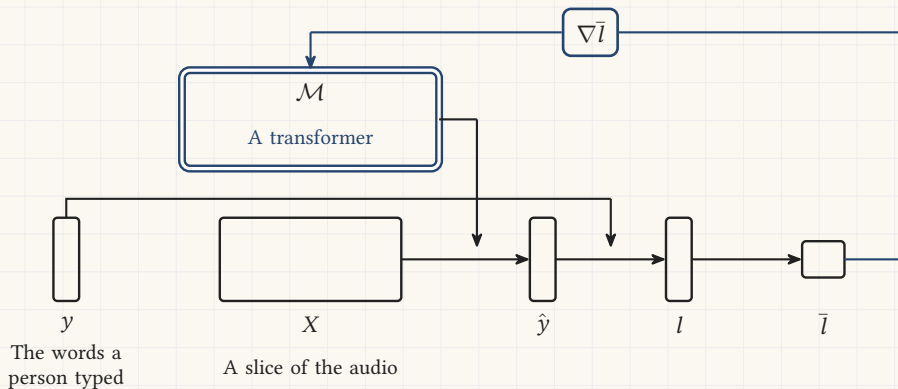
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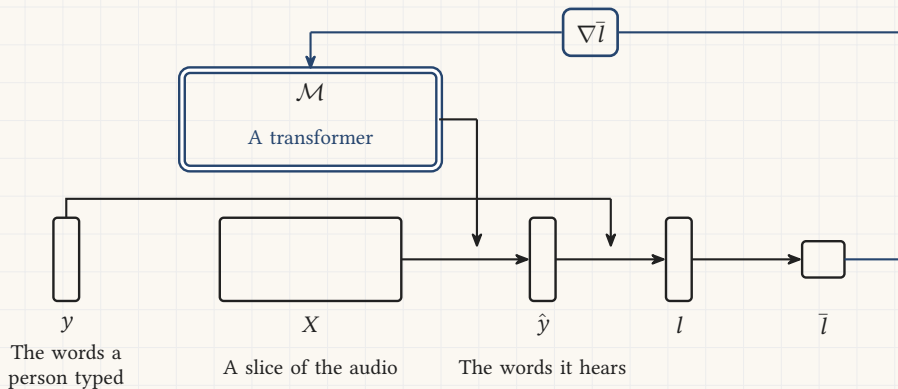
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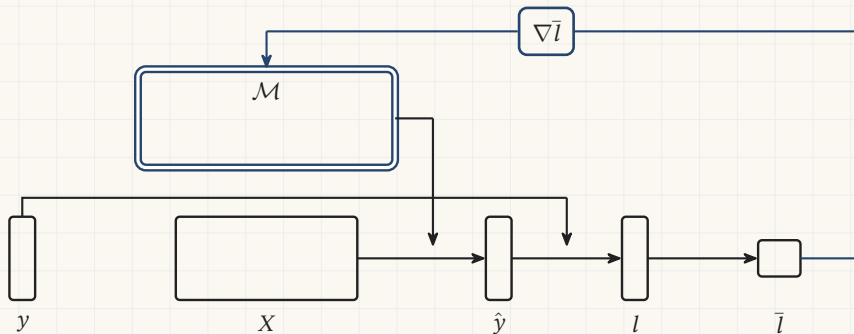
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Ranking and recommendation

Every impression is a row: what was shown, to whom, and whether they took it.

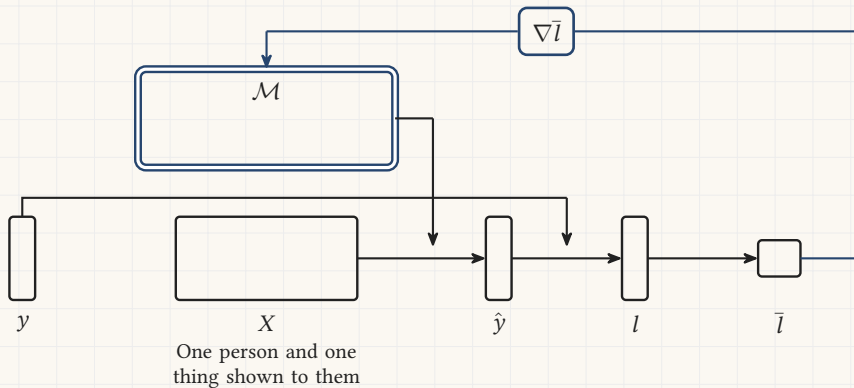
The system that picks the next video. Covington, Adams and Sargin, RecSys 2016.



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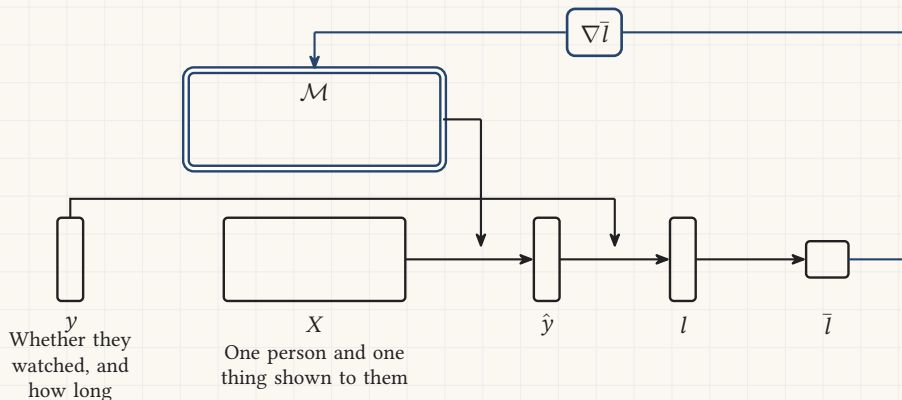
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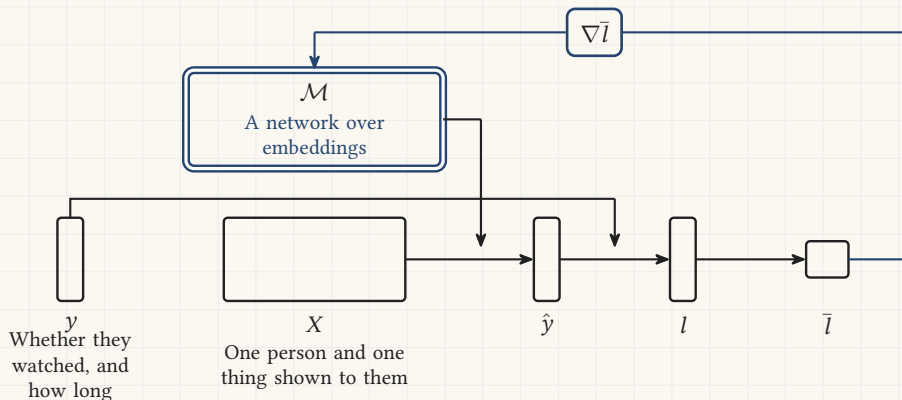
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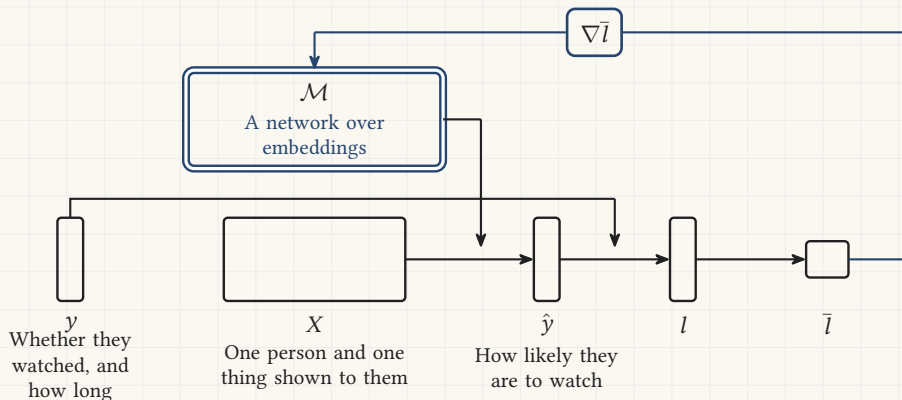
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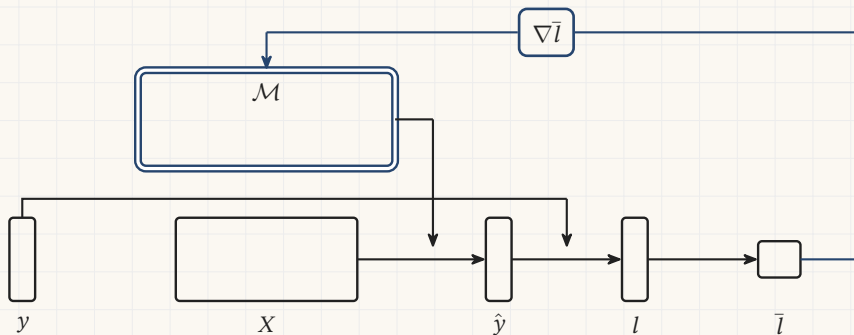
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Fraud and credit decisions

Every settled transaction is a row, and its label arrives weeks later with the chargeback.

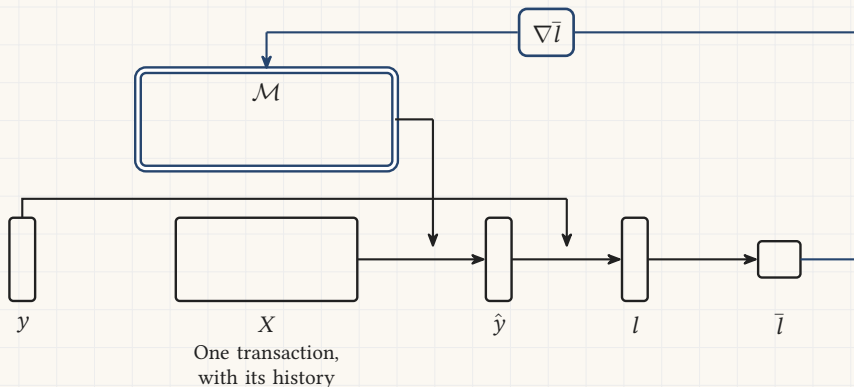
Dal Pozzolo and others, 2018, with Worldline. The tree ensemble is fitted to the gradient too.



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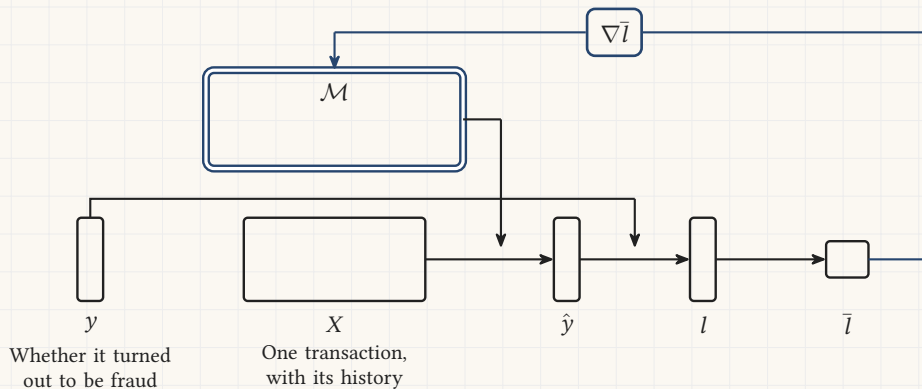
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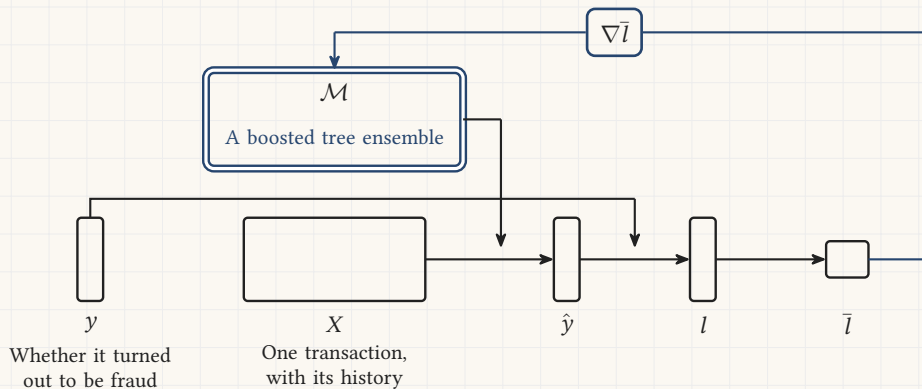
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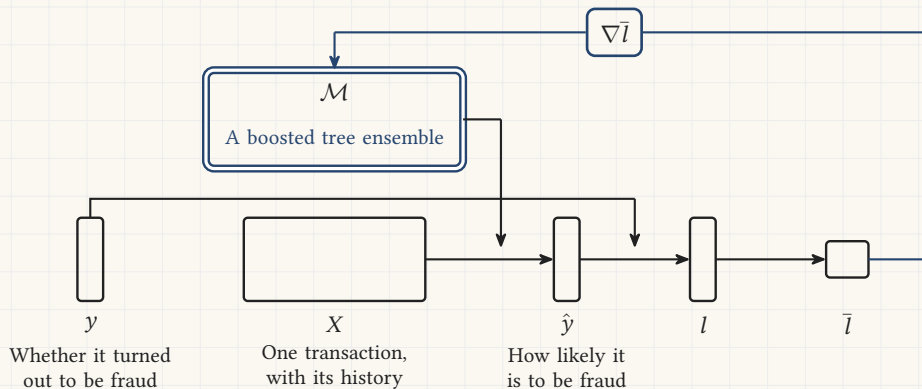
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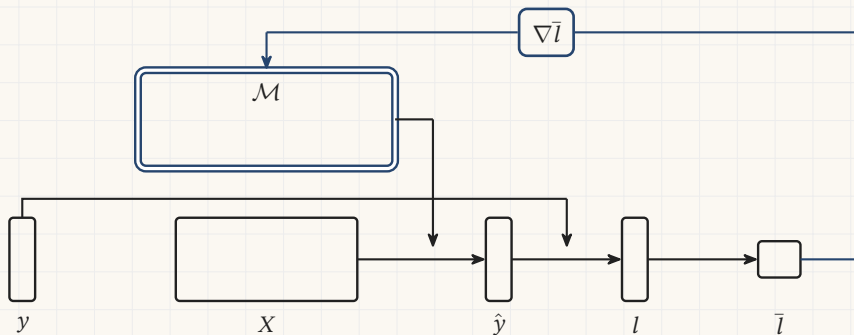
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Screening for diabetic eye disease

In 2018 a regulator cleared a machine to read these photographs with no doctor behind it.

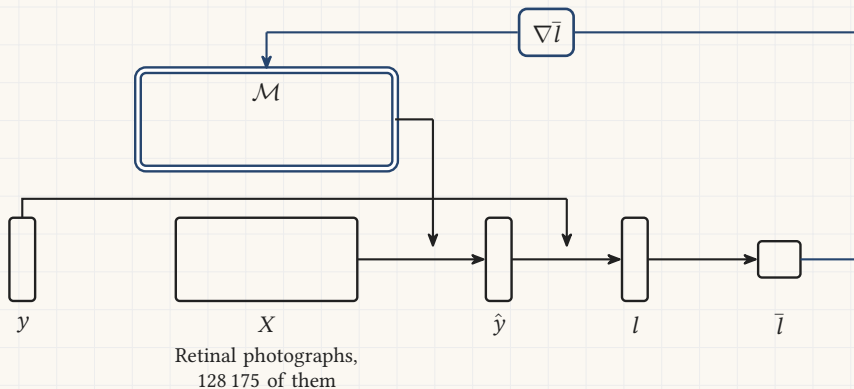
128 175 photographs, graded 3 to 7 times by 54 ophthalmologists. Gulshan and others, *JAMA*, 2016.



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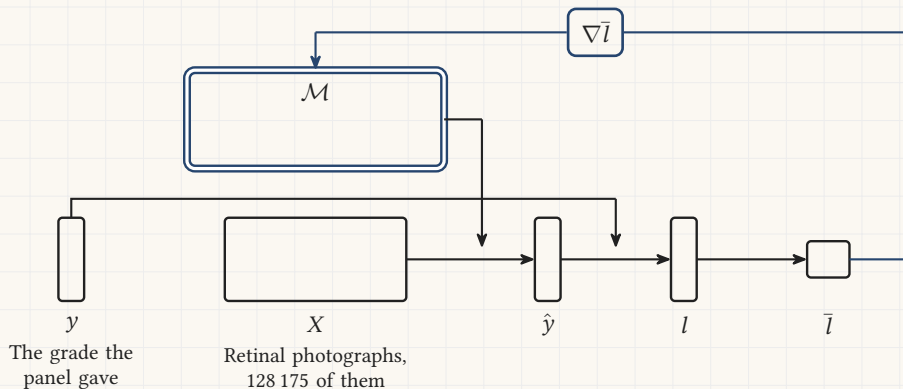
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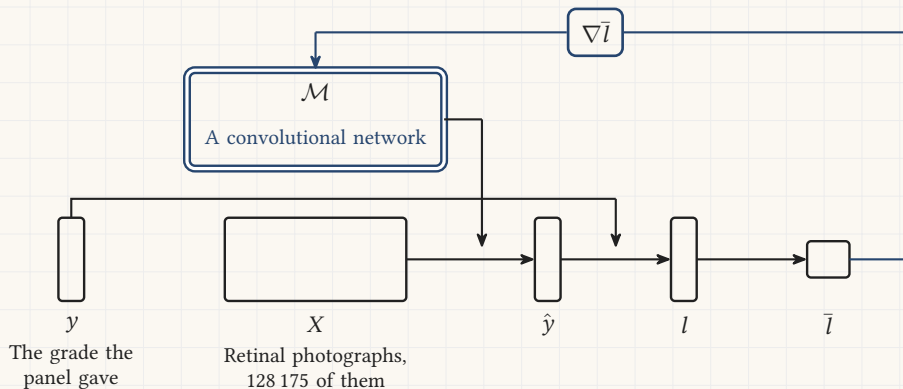
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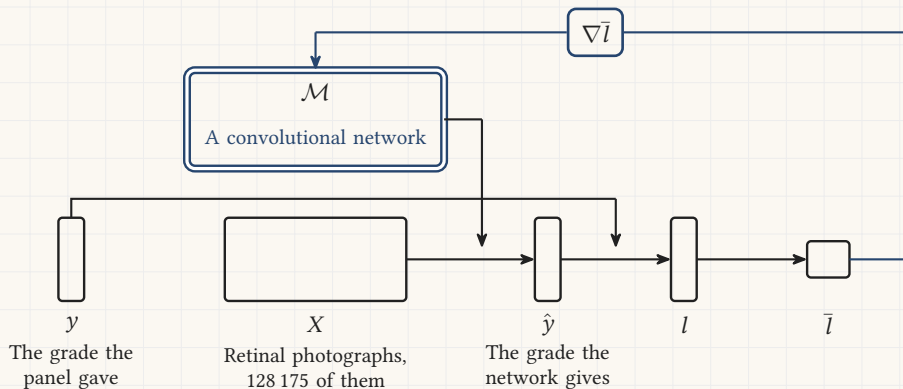
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Four outcomes

The trial behind that clearance: 819 people at ten primary care clinics.

Abràmoff and others, *npj Digital Medicine*, 2018. Counts from its figure 1.

		<i>What the experts found</i>	
		Has the disease	Does not
<i>What the machine said</i>	Refer		
	All clear		

Four outcomes

The trial behind that clearance: 819 people at ten primary care clinics.

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		What the experts found	
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		Has the disease 198 people	Does not 621 people
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	All clear	25 Missed	556 Rightly cleared

Four outcomes

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		<i>What the experts found</i>	
		Has the disease 198 people	Does not 621 people
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	All clear	25 Missed False negative	556 Rightly cleared True negative

Precision and recall

Two questions to ask of any machine that answers yes or no.

Medicine calls recall sensitivity. Abràmoff and others, *npj Digital Medicine*, 2018.

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Recall: of the 198 who had it,
it caught 173, about 87 in 100

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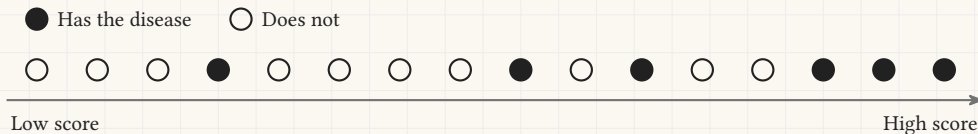
Precision: of the 238 it referred, 173 had it, about 73 in 100

Recall: of the 198 who had it, it caught 173, about 87 in 100

The threshold

The machine gives every case a score. People choose the score from which it says yes.

A schematic. In the trial the threshold was fixed before the first patient.



A schematic. In the trial the threshold was fixed before the first patient.

● Has the disease ○ Does not

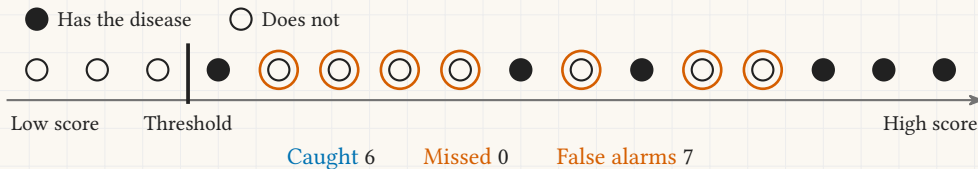
Low score Threshold High score

Caught 4 Missed 2 False alarms 2

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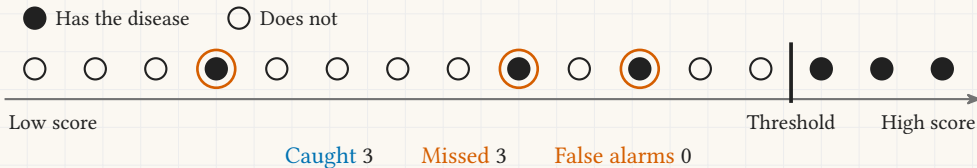
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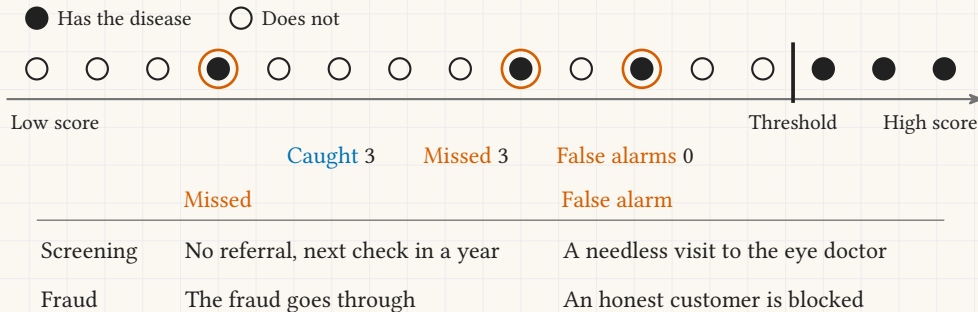
A schematic. In the trial the threshold was fixed before the first patient.

[illegible]

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A schematic. In the trial the threshold was fixed before the first patient.



References: Vapnik and the language models

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From LLMs to reasoning models

Agentic AI, part 1

Carlos Cotrini

UZH CAS Gamechanger AI

26 September 2026

1

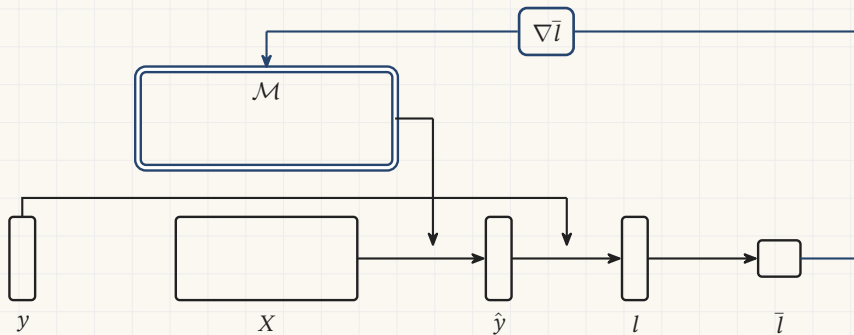
Section 1

How a language model is trained

The loop, with the model as one box

The picture from weekend 1, with the network as one box: the model $\mathcal{M}$.

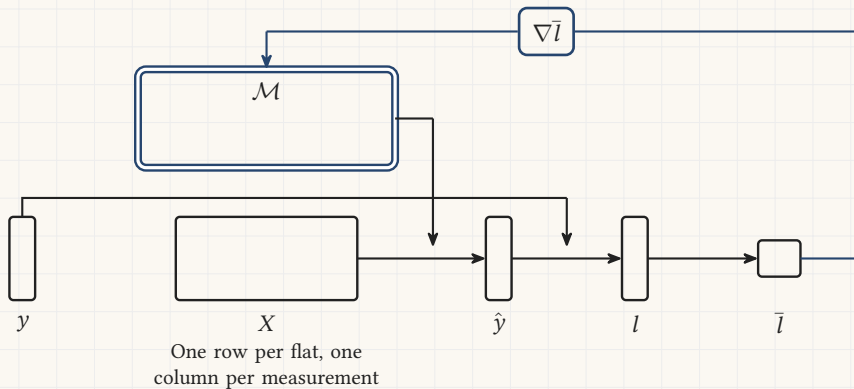
The five flats and the network are the ones from weekend 1.



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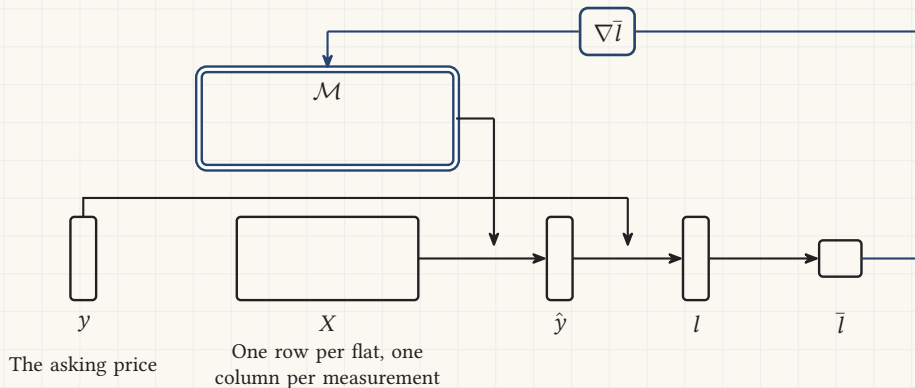
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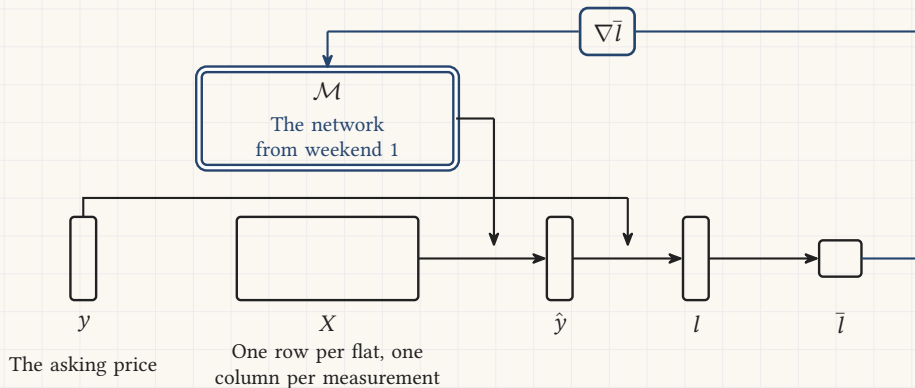
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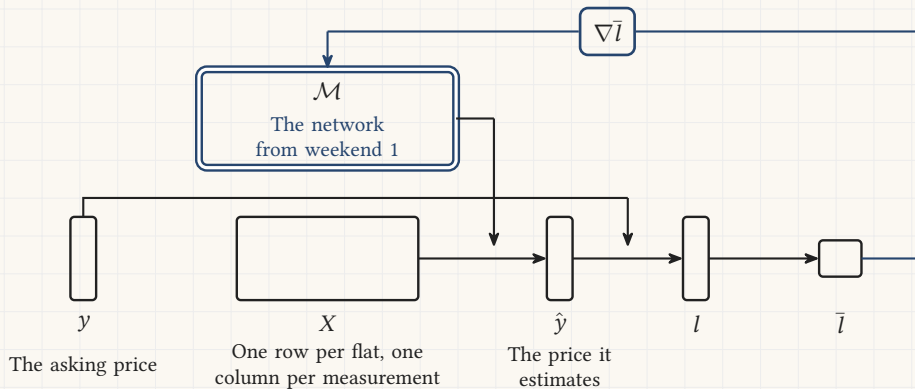
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The loop, with the model as one box

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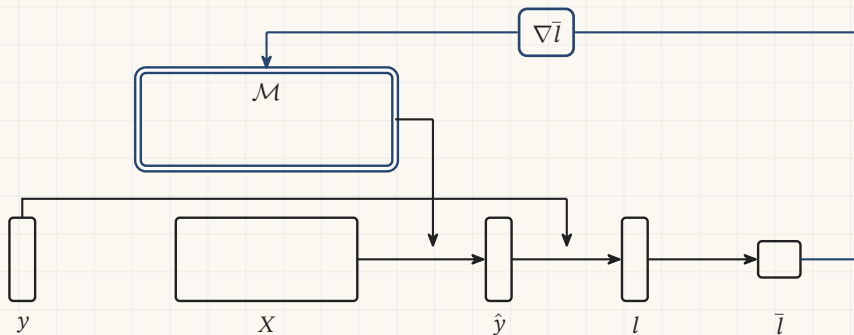
The five flats and the network are the ones from weekend 1.



Large language models

A row is text with its next token cut off. Every position of every document gives one.

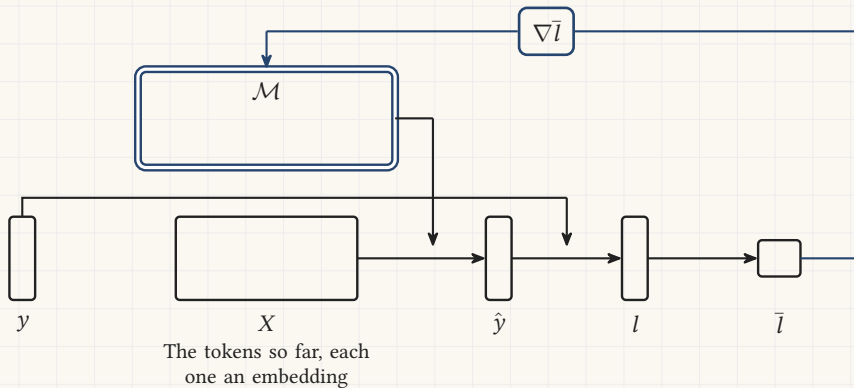
The box holds a transformer. Vaswani and others, *Attention is all you need*, 2017.



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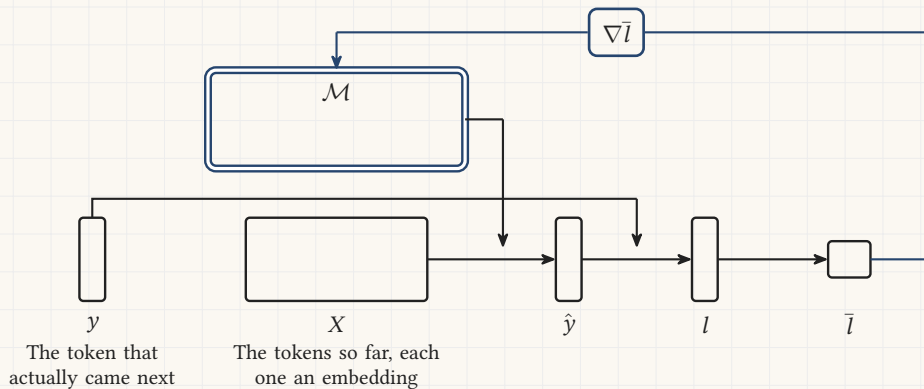
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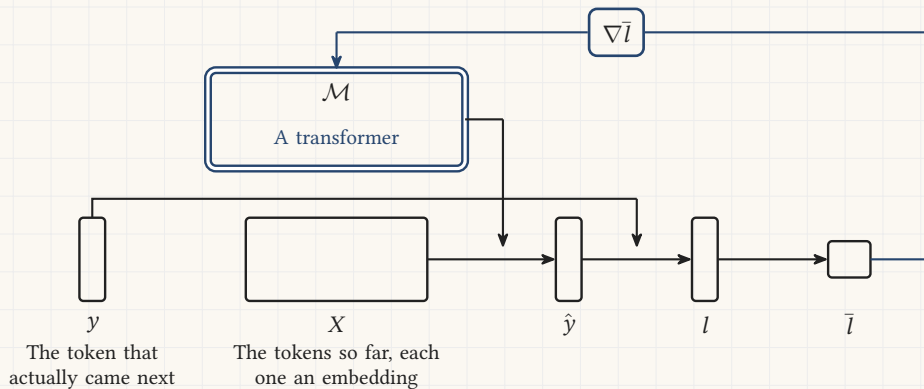
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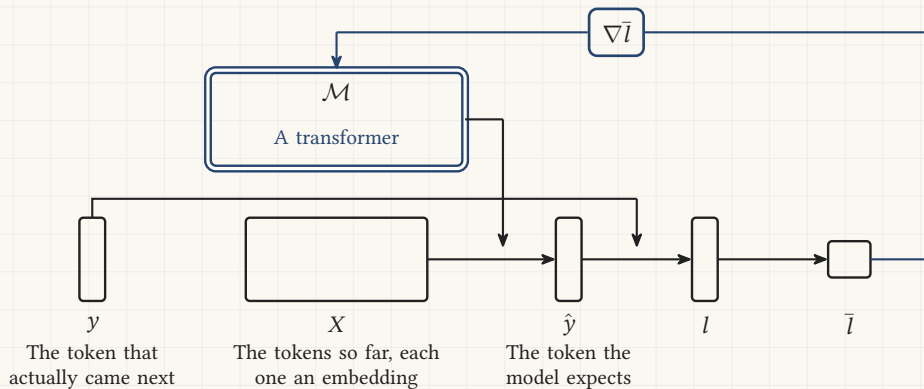
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Large language models

A row is text with its next token cut off. Every position of every document gives one.

The box holds a transformer. Vaswani and others, *Attention is all you need*, 2017.



One row for every position

	feat 1	feat 2	...	feat d
once				
upon				
a				
time				
there				
was				

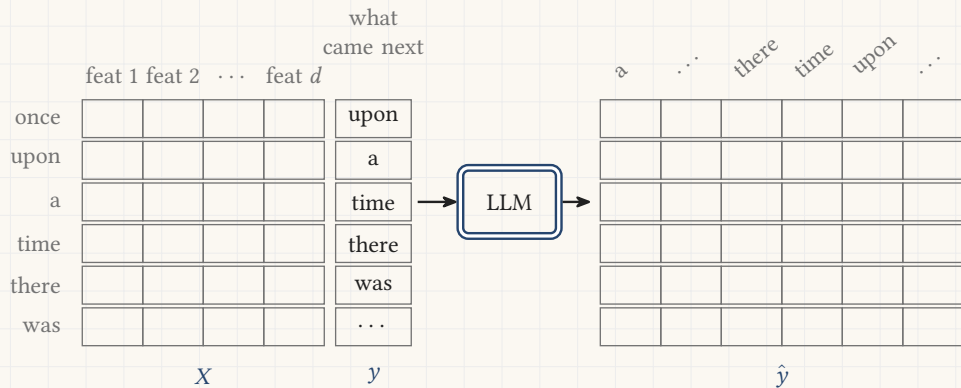
X

One row for every position

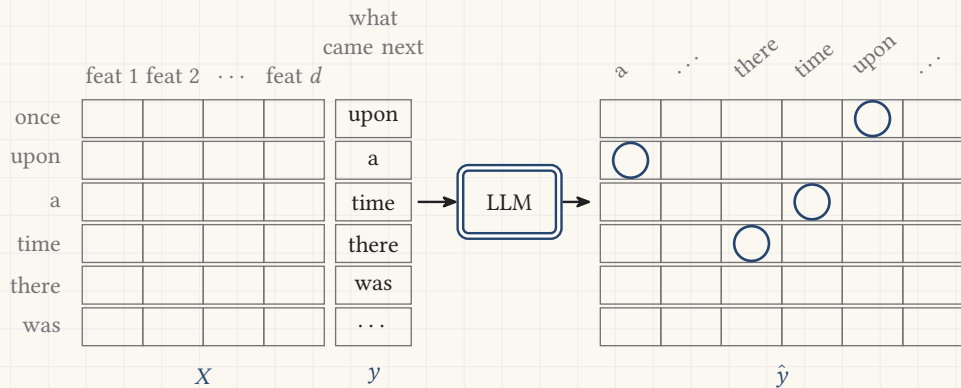
	what came next			
	feat 1	feat 2	...	feat d
once				upon
upon				a
a				time
time				there
there				was
was				...
	X			
	y			



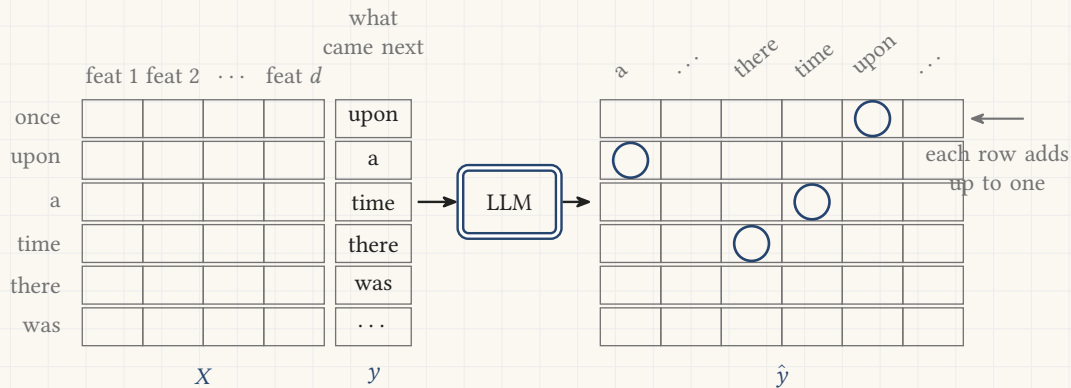
One row for every position



One row for every position



One row for every position



Reading that matrix

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Every row One position of the context. The sentence has six words, so $\hat{y}$ has six rows.

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Every column One token of the vocabulary. Every word in the dictionary gets a column, whether or not it appears in the sentence.

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Every entry $\hat{y}_{ij}$ is how strongly the model believes the word at position i should be word j .

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$$\sum_j \hat{y}_{ij} = 1 \quad \text{for every row } i$$

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$$\sum_j \hat{y}_{ij} = 1 \quad \text{for every row } i$$

So the output is a whole table of beliefs, one for every position at once. Predicting the next word is reading a single row of it.

Scoring the whole table with one number

Read off, in each row, the entry for the word that actually came next.

$$\tilde{\ell} = -\hat{y}_{1, \text{"upon"}}$$

Scoring the whole table with one number

Read off, in each row, the entry for the word that actually came next.

$$\tilde{\ell} = -\hat{y}_{1, \text{"upon"}} - \hat{y}_{2, \text{"a"}}$$

Scoring the whole table with one number

Read off, in each row, the entry for the word that actually came next.

$$\tilde{\ell} = -\hat{y}_{1, \text{"upon"}} - \hat{y}_{2, \text{"a"}} - \hat{y}_{3, \text{"time"}} - \dots$$

Scoring the whole table with one number

Read off, in each row, the entry for the word that actually came next.

$$\tilde{\ell} = -\hat{y}_{1, \text{"upon"}} - \hat{y}_{2, \text{"a"}} - \hat{y}_{3, \text{"time"}} - \dots$$

Both versions are smallest when the model puts all of its weight on the word that actually came next.

Scoring the whole table with one number

Read off, in each row, the entry for the word that actually came next.

$$\ell = -\log \hat{y}_{1, \text{"upon"}} - \log \hat{y}_{2, \text{"a"}} - \log \hat{y}_{3, \text{"time"}} - \dots$$

Both versions are smallest when the model puts all of its weight on the word that actually came next.

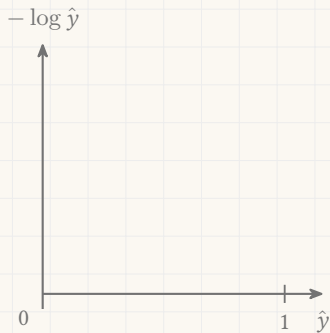
Scoring the whole table with one number

Read off, in each row, the entry for the word that actually came next.

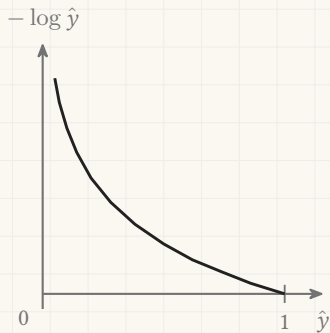
$$\ell = -\log \hat{y}_{1, \text{"upon"}} - \log \hat{y}_{2, \text{"a"}} - \log \hat{y}_{3, \text{"time"}} - \dots$$

Both versions are smallest when the model puts all of its weight on the word that actually came next.
They differ in how hard they push when it does not.

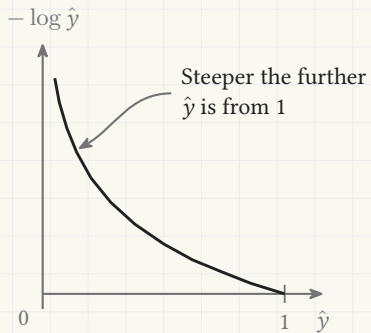
Why the logarithm



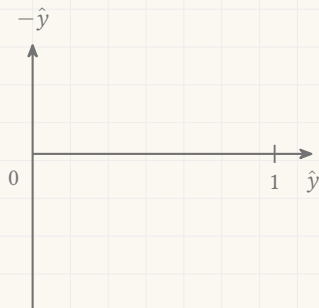
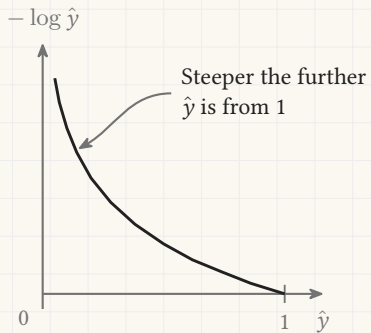
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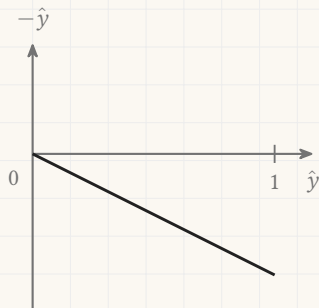
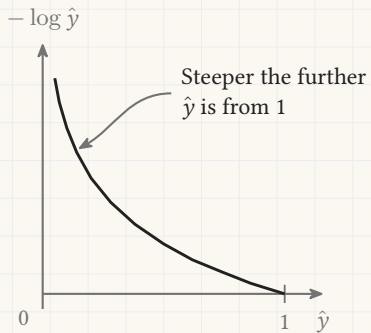
Why the logarithm



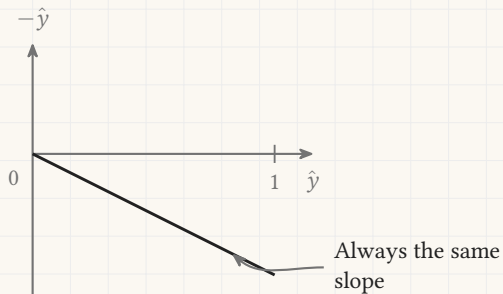
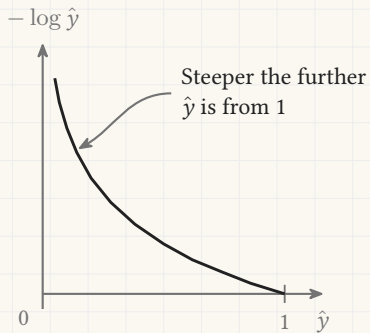
Why the logarithm



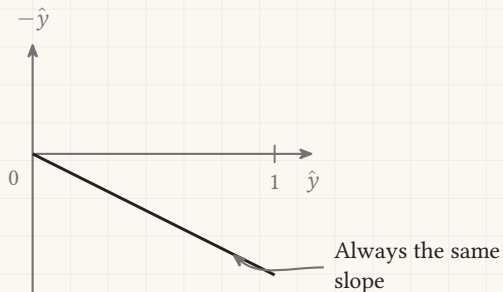
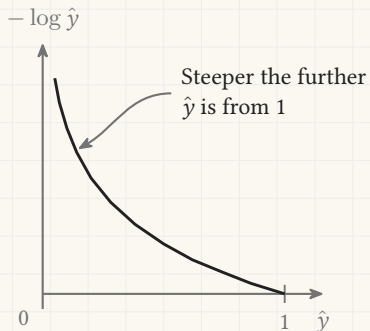
Why the logarithm



Why the logarithm



Why the logarithm



The slope is the gradient. A constant one pushes just as gently on a word the model gave no chance as on one it nearly had right.

2

Section 2

How a language model writes text

Greedy decoding and sampling

To write, the model reads one row: the row after the text so far.

Six words stand in for the dictionary. The numbers are made up.

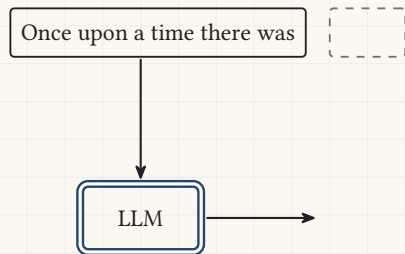
Once upon a time there was



Greedy decoding and sampling

To write, the model reads one row: the row after the text so far.

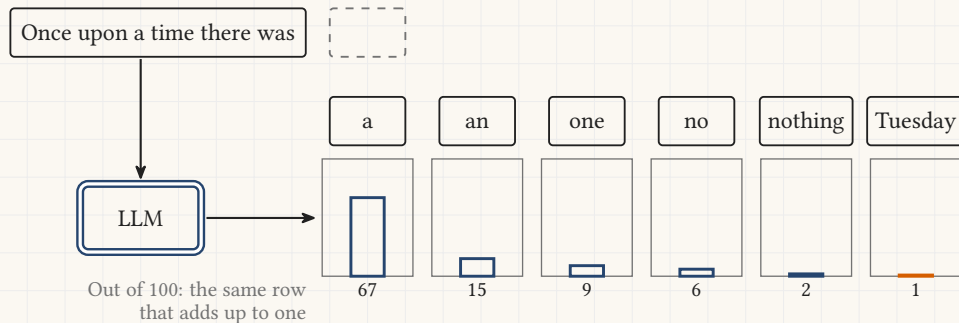
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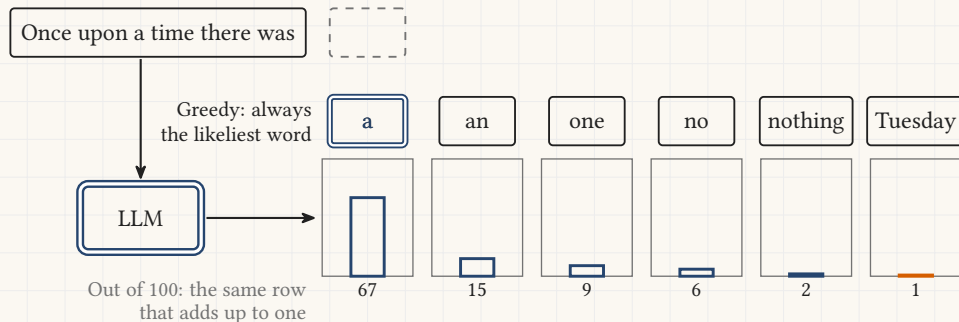
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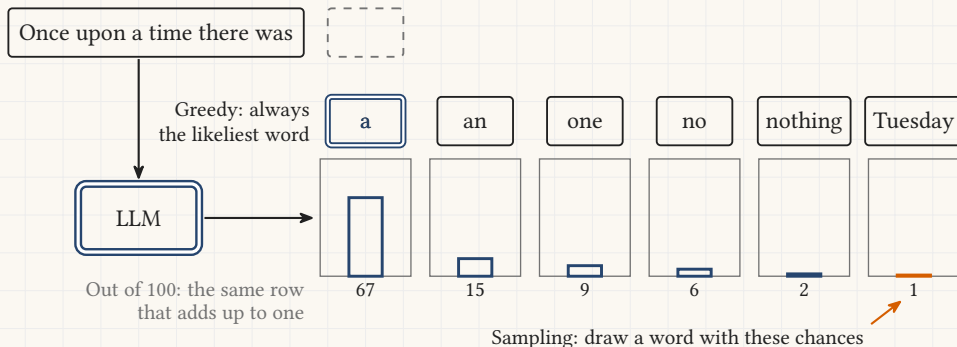
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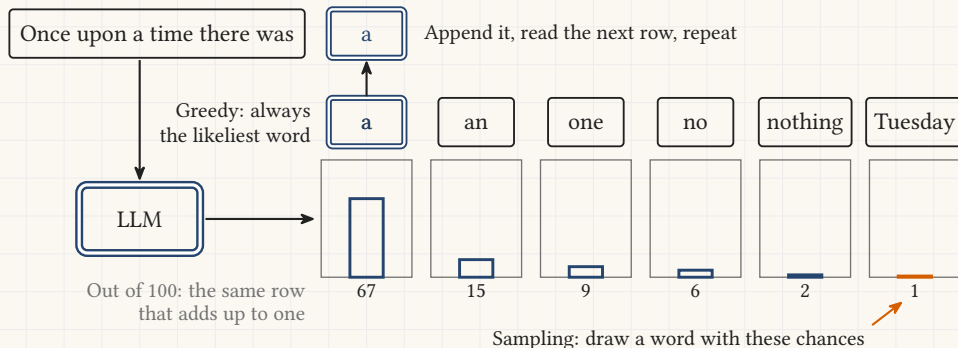
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Temperature

Behind the row is a score for every word. Every score is divided by the temperature first.

Jurafsky and Martin, *Speech and Language Processing*, 3rd edition draft, 2026, section 7.6.

	a	an	one	no	nothing	Tuesday
Score	4	2.5	2	1.5	0.5	0

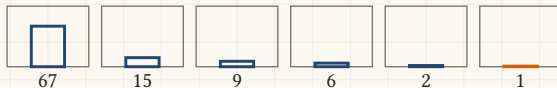
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Temperature 1
Scores as they are

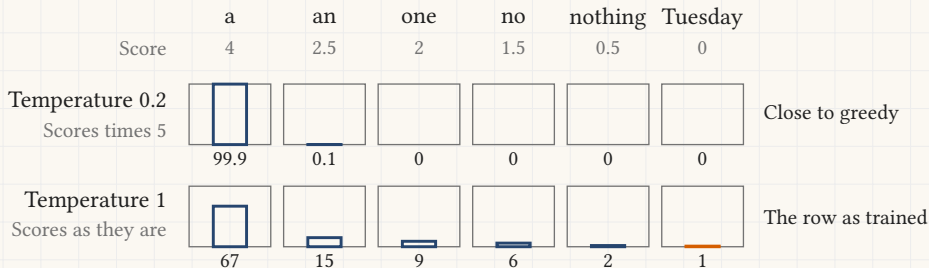


The row as trained

Temperature

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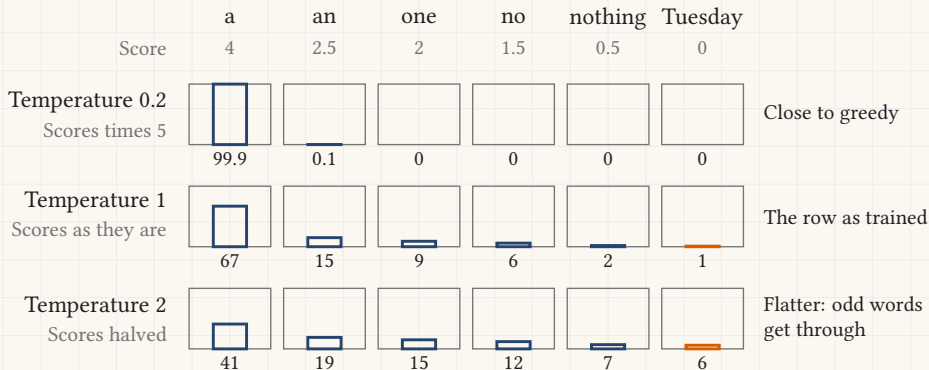
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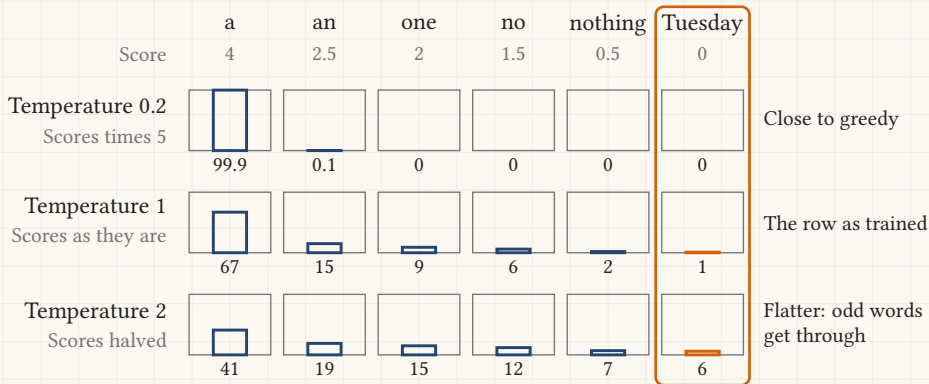
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Jurafsky and Martin, *Speech and Language Processing*, 3rd edition draft, 2026, section 7.6.



Temperature in practice

A word, once written, stays. Every later row is read after it.

Holtzman and others, ICLR 2020. Zhang and others, *How language model hallucinations can snowball*, ICML 2024.

Once upon a time there was

Tuesday

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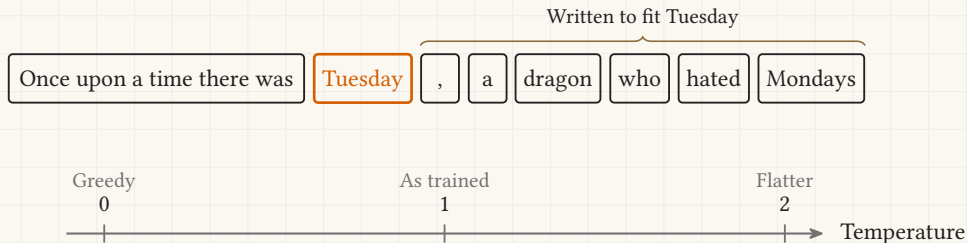
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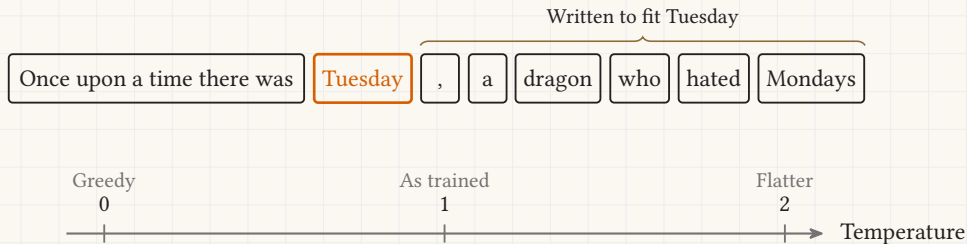
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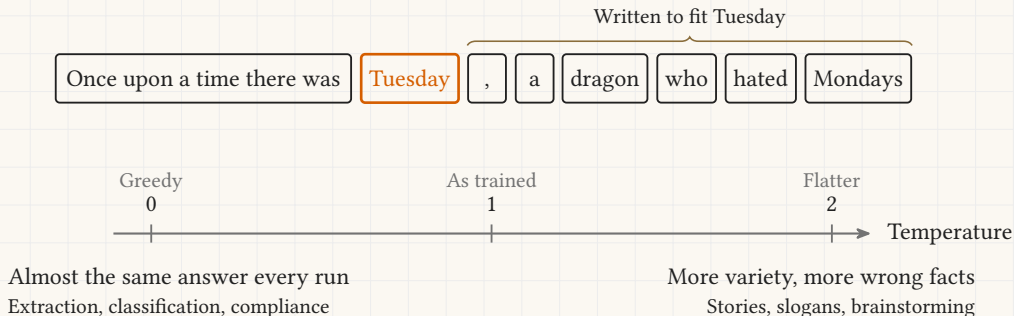


Almost the same answer every run
Extraction, classification, compliance

Temperature in practice

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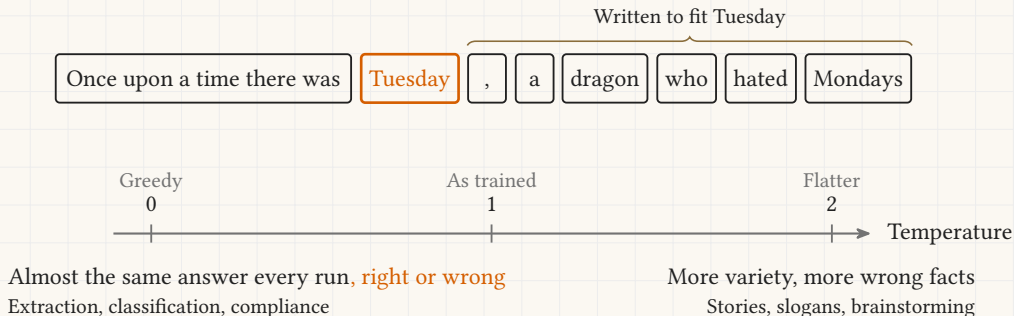
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Holtzman and others, ICLR 2020. Zhang and others, *How language model hallucinations can snowball*, ICML 2024.



3

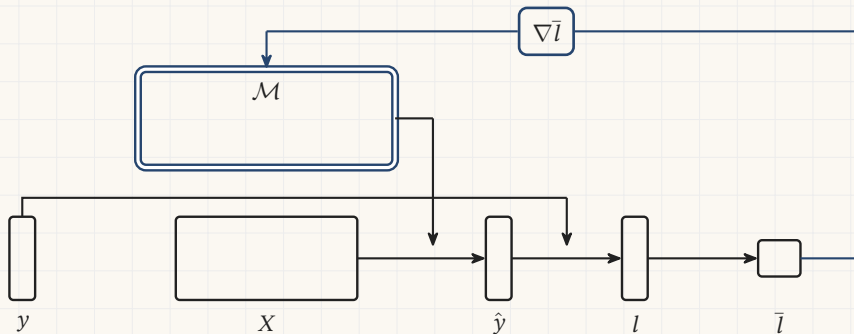
Section 3

How a language model becomes a chatbot

Instruction tuning

GPT-3 was finished in 2020 and it completed text. A smaller second pass made it answer.

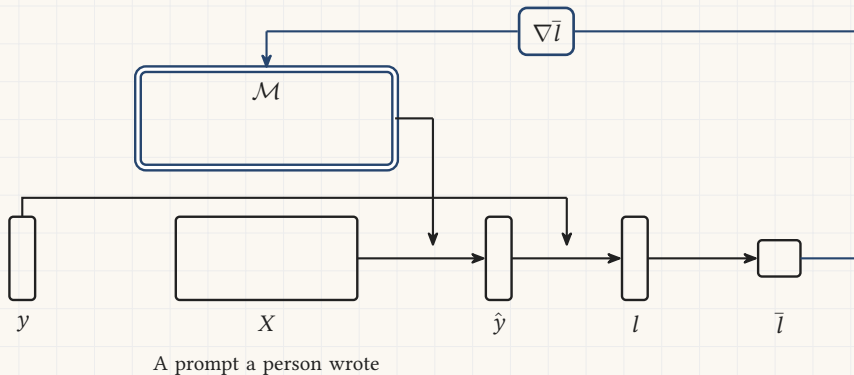
12 725 training prompts, and its replies were preferred to a model 100 times larger. Ouyang and others, 2022.



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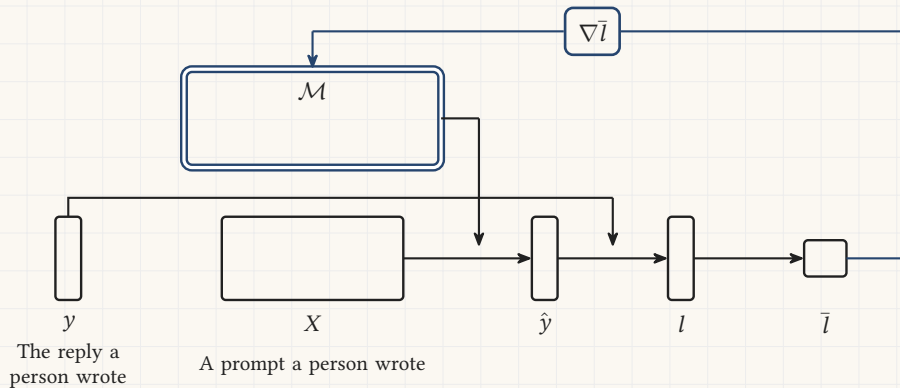
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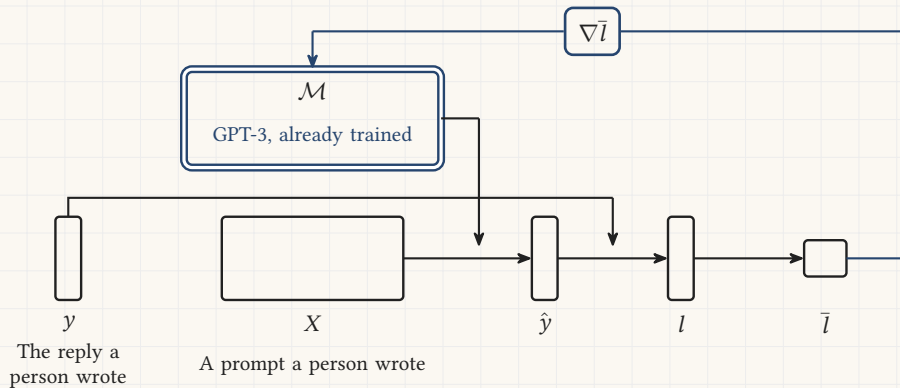
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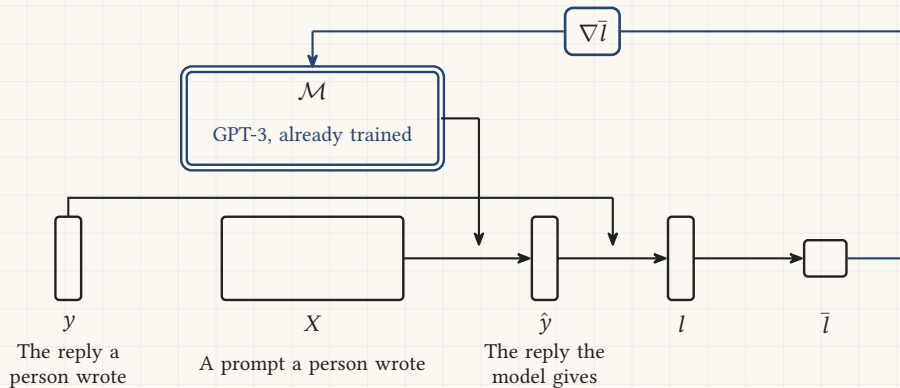
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The instruction tuning data

About 40 contractors, hired on Upwork and through Scale AI, wrote the replies and did the ranking.

Example from figure 2, counts from table 6. Ouyang and others, 2022.

One demonstration

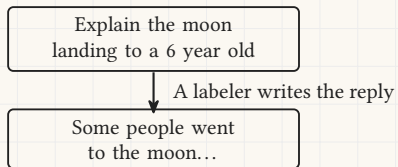
Explain the moon
landing to a 6 year old

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One demonstration



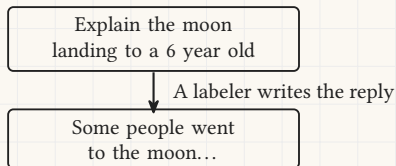
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One demonstration

12 725 prompts, most of them written by the labelers



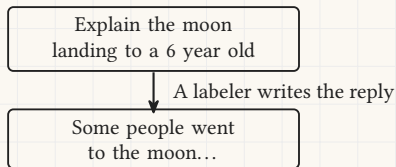
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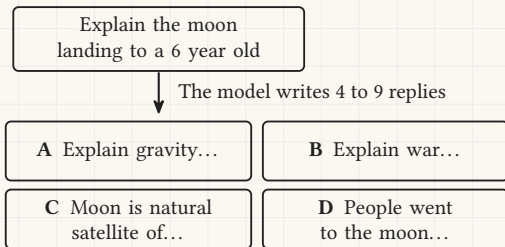
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One demonstration

12 725 prompts, most of them written by the labelers



One comparison



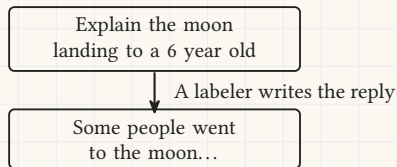
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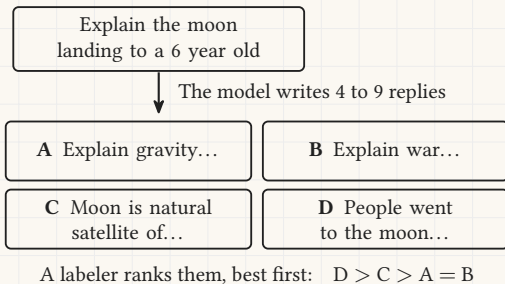
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One demonstration

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One comparison



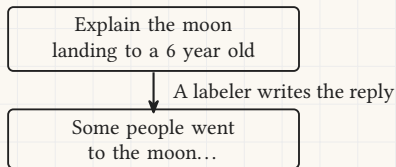
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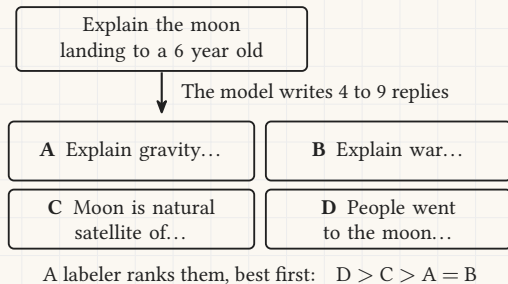
One demonstration

12 725 prompts, most of them written by the labelers



One comparison

33 207 prompts, most of them sent by users of the API



4

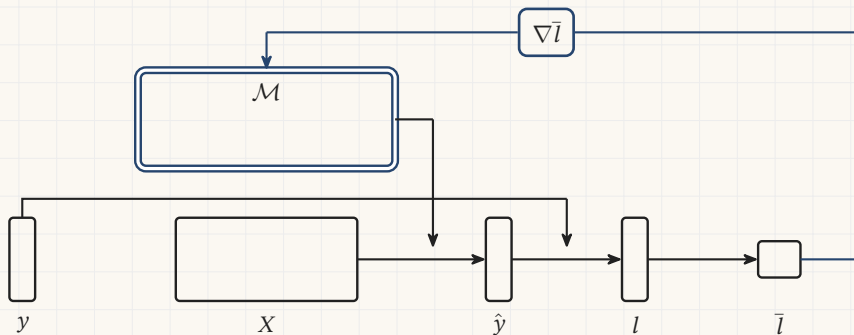
Section 4

How a chatbot becomes a reasoning model

Reasoning models

Start from a trained language model and keep going, on problems a machine can mark.

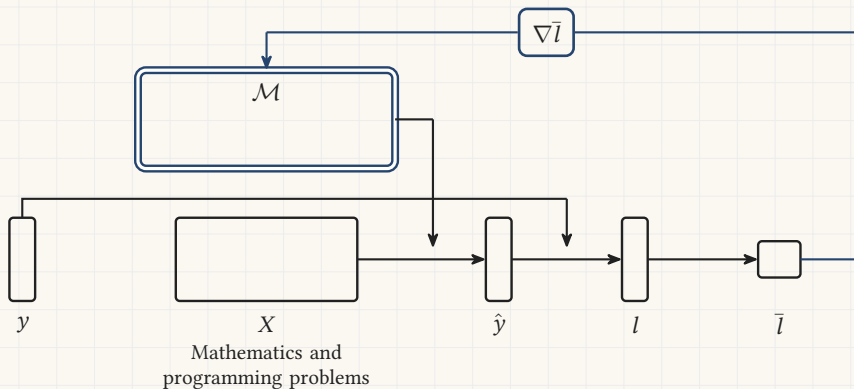
The verdict is a marker rather than a subtraction. DeepSeek-R1, *Nature* 645, 633 (2025).



Reasoning models

Start from a trained language model and keep going, on problems a machine can mark.

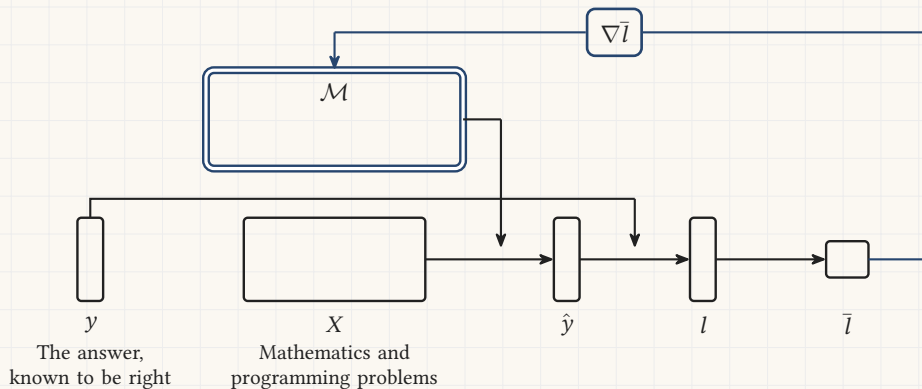
The verdict is a marker rather than a subtraction. DeepSeek-R1, *Nature* 645, 633 (2025).



Reasoning models

Start from a trained language model and keep going, on problems a machine can mark.

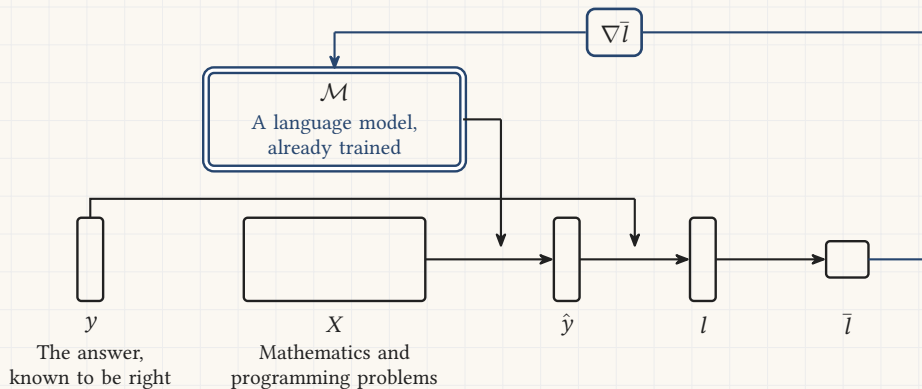
The verdict is a marker rather than a subtraction. DeepSeek-R1, *Nature* 645, 633 (2025).



Reasoning models

Start from a trained language model and keep going, on problems a machine can mark.

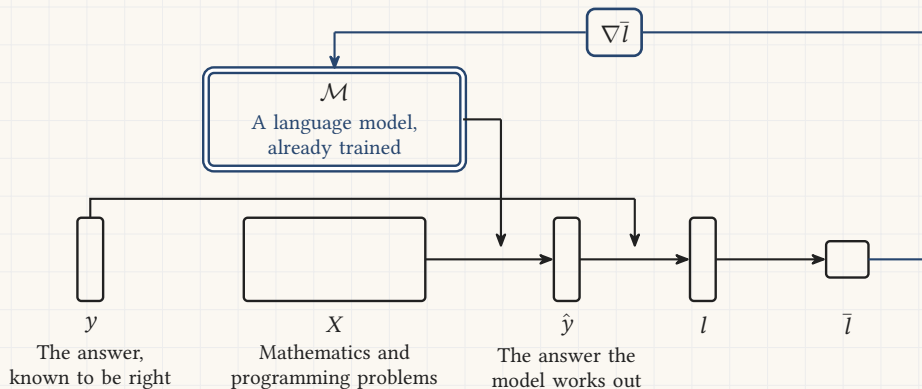
The verdict is a marker rather than a subtraction. DeepSeek-R1, *Nature* 645, 633 (2025).



Reasoning models

Start from a trained language model and keep going, on problems a machine can mark.

The verdict is a marker rather than a subtraction. DeepSeek-R1, *Nature* 645, 633 (2025).



5

Section 5

Reinforcement learning

A language model as a Markov decision process

Writing an answer is a chain of decisions, one token at a time.

Words stand in for tokens. Liu and others, *Understanding R1-Zero-like training*, 2025.

What is 12 times 7?

12

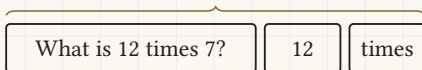
times

A language model as a Markov decision process

Writing an answer is a chain of decisions, one token at a time.

Words stand in for tokens. Liu and others, *Understanding R1-Zero-like training*, 2025.

The state: the prompt and the tokens written so far

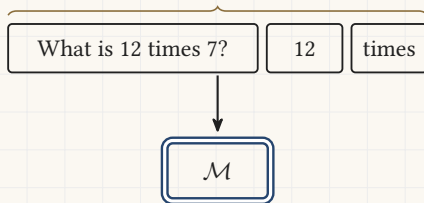


A language model as a Markov decision process

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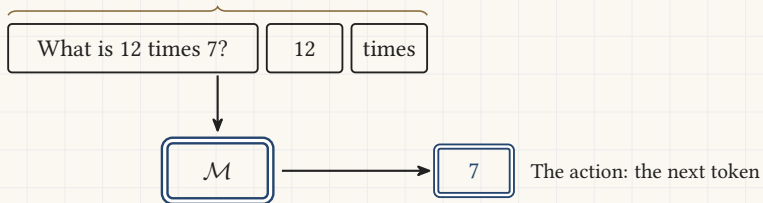


A language model as a Markov decision process

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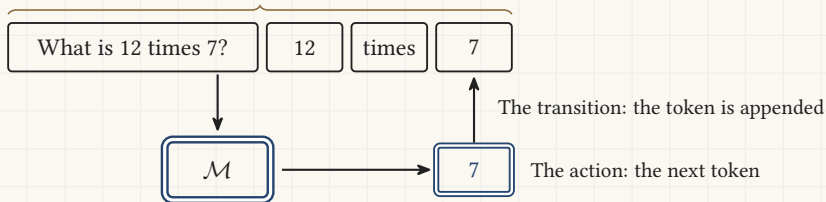


A language model as a Markov decision process

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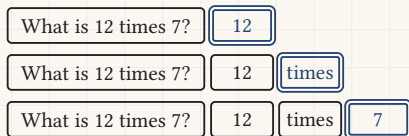
The state: the prompt and the tokens written so far



The end of the episode, and the reward

The model writes until it chooses the end-of-answer token. Then a program checks the answer.

DeepSeek-R1 was trained with rewards like these. DeepSeek-AI, *Nature* 645, 633 (2025).



The end of the episode, and the reward

The model writes until it chooses the end-of-answer token. Then a program checks the answer.

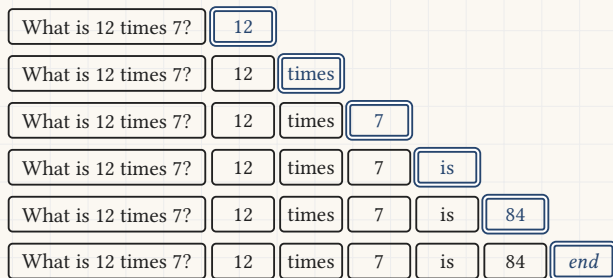
DeepSeek-R1 was trained with rewards like these. DeepSeek-AI, *Nature* 645, 633 (2025).

What is 12 times 7?	12				
What is 12 times 7?	12	times			
What is 12 times 7?	12	times	7		
What is 12 times 7?	12	times	7	is	
What is 12 times 7?	12	times	7	is	84

The end of the episode, and the reward

The model writes until it chooses the end-of-answer token. Then a program checks the answer.

DeepSeek-R1 was trained with rewards like these. DeepSeek-AI, *Nature* 645, 633 (2025).



The end-of-answer token: the episode ends

The end of the episode, and the reward

The model writes until it chooses the end-of-answer token. Then a program checks the answer.

DeepSeek-R1 was trained with rewards like these. DeepSeek-AI, *Nature* 645, 633 (2025).

						Reward
What is 12 times 7?	12					0
What is 12 times 7?	12	times				0
What is 12 times 7?	12	times	7			0
What is 12 times 7?	12	times	7	is		0
What is 12 times 7?	12	times	7	is	84	0
What is 12 times 7?	12	times	7	is	84	end

0 while the answer is still being written

The end-of-answer token: the episode ends

The end of the episode, and the reward

The model writes until it chooses the end-of-answer token. Then a program checks the answer.

DeepSeek-R1 was trained with rewards like these. DeepSeek-AI, *Nature* 645, 633 (2025).

						Reward
What is 12 times 7?	12					0
What is 12 times 7?	12	times				0
What is 12 times 7?	12	times	7			0
What is 12 times 7?	12	times	7	is		0
What is 12 times 7?	12	times	7	is	84	0
What is 12 times 7?	12	times	7	is	84	1
					end	0
The end-of-answer token: the episode ends						

0 while the answer is still being written

1 at the end: 84 is right.
A wrong answer earns 0

The end of the episode, and the reward

The model writes until it chooses the end-of-answer token. Then a program checks the answer.

DeepSeek-R1 was trained with rewards like these. DeepSeek-AI, *Nature* 645, 633 (2025).

						Reward	
What is 12 times 7?	12					0	
What is 12 times 7?	12	times				0	
What is 12 times 7?	12	times	7			0	
What is 12 times 7?	12	times	7	is		0	
What is 12 times 7?	12	times	7	is	84	0	
What is 12 times 7?	12	times	7	is	84	end	1
The end-of-answer token: the episode ends						1	0 while the answer is still being written
						1	1 at the end: 84 is right. A wrong answer earns 0
						1	The sum of the rewards

Gradient descent, from weekend 1

Weekend 1 made a loss smaller by stepping against its tilt.

L is the loss, for example the average squared miss over the houses.

$w \leftarrow w$
The weights

Gradient descent, from weekend 1

Weekend 1 made a loss smaller by stepping against its tilt.

L is the loss, for example the average squared miss over the houses.

$$\underbrace{w}_{\text{The weights}} \leftarrow w - \underbrace{\alpha}_{\text{Minus a small step}}$$

Gradient descent, from weekend 1

Weekend 1 made a loss smaller by stepping against its tilt.

L is the loss, for example the average squared miss over the houses.

$$\underbrace{w}_{\text{The weights}} \leftarrow w \underbrace{- \alpha}_{\text{Minus a small step}} \underbrace{\nabla}_{\text{The direction that increases}}$$

Gradient descent, from weekend 1

Weekend 1 made a loss smaller by stepping against its tilt.

L is the loss, for example the average squared miss over the houses.

$$\underbrace{w}_{\text{The weights}} \leftarrow \underbrace{w}_{\text{Minus a small step}} - \underbrace{\alpha}_{\text{The direction that increases}} \underbrace{L(w)}_{\text{The loss}}$$

Gradient ascent on the sum of rewards

Here the step goes with the tilt, to make the expected sum of rewards larger.

Policy gradient methods. Sutton and Barto, *Reinforcement Learning*, 2nd edition, 2018, chapter 13.

$w \leftarrow w$
The weights

Gradient ascent on the sum of rewards

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$$\underbrace{w}_{\text{The weights}} \leftarrow w + \underbrace{\alpha}_{\text{Plus a small step}}$$

Gradient ascent on the sum of rewards

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$$\underbrace{w}_{\text{The weights}} \leftarrow \underbrace{w + \alpha}_{\text{Plus a small step}} \underbrace{\nabla}_{\text{The direction that increases}}$$

Gradient ascent on the sum of rewards

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Policy gradient methods. Sutton and Barto, *Reinforcement Learning*, 2nd edition, 2018, chapter 13.

$$\underbrace{w}_{\text{The weights}} \leftarrow \underbrace{w}_{\text{Plus a small step}} + \underbrace{\alpha}_{\text{The direction that increases}} \underbrace{\mathbb{E}[r_1 + \dots + r_T]}_{\text{The expected sum of rewards}}$$

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Agentic AI, part 2

Agents

Carlos Cotrini

UZH CAS Gamechanger AI

26 September 2026

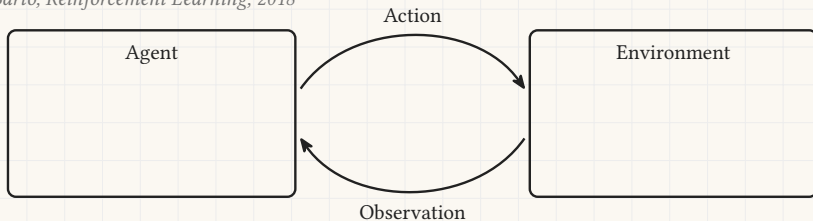
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Section 1

Agents and the loop

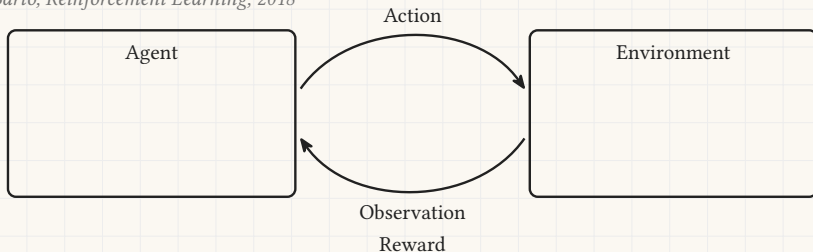
The agent and its environment

Sutton and Barto, Reinforcement Learning, 2018



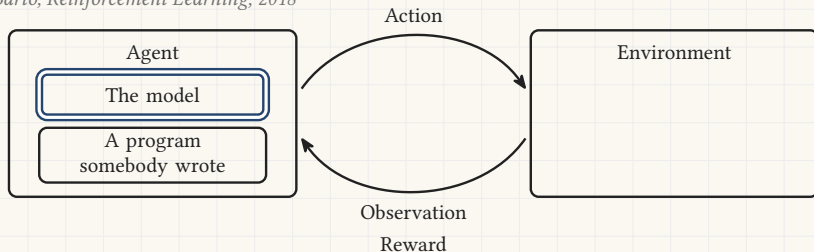
The agent and its environment

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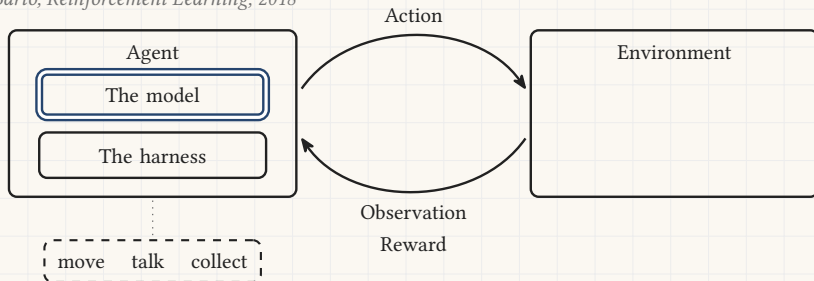
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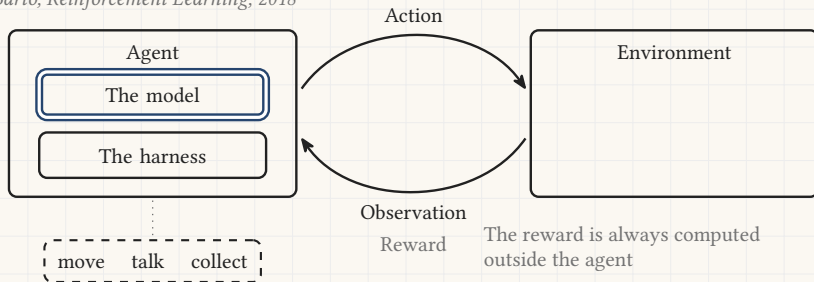
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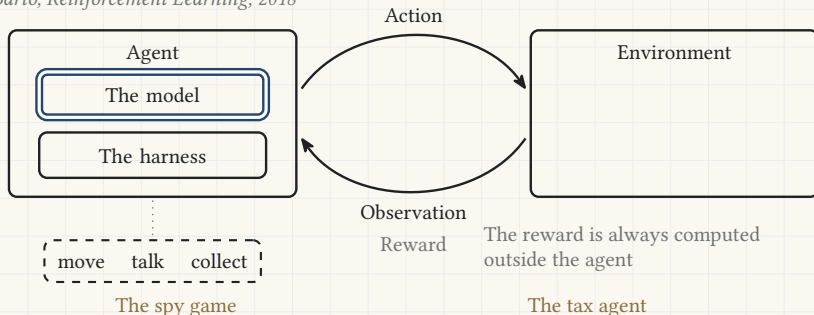
The agent and its environment

Sutton and Barto, Reinforcement Learning, 2018



The agent and its environment

Sutton and Barto, Reinforcement Learning, 2018



Environment

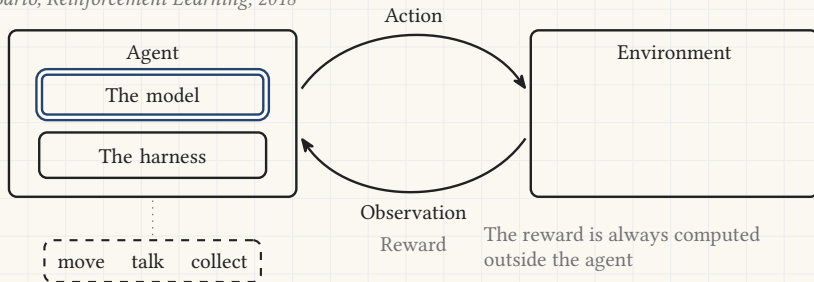
Observations

Actions

Reward

The agent and its environment

Sutton and Barto, Reinforcement Learning, 2018



The reward is always computed outside the agent

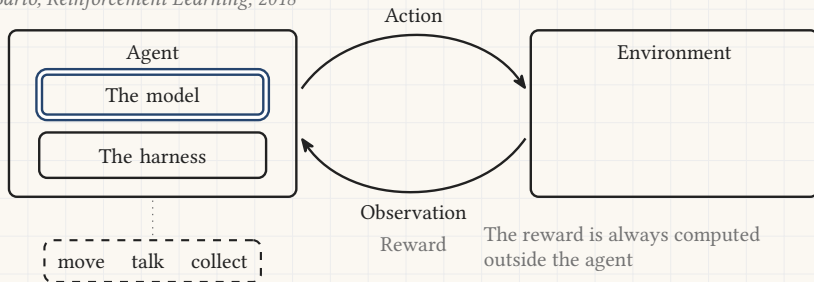
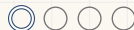
The spy game

The tax agent

Environment	The 3 by 3 base
Observations	What the game prints after each move
Actions	Three tools, and one scan the harness runs
Reward	None is computed

The agent and its environment

Sutton and Barto, Reinforcement Learning, 2018



The reward is always computed outside the agent

The spy game

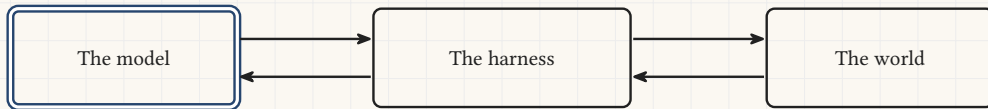
Environment	The 3 by 3 base
Observations	What the game prints after each move
Actions	Three tools, and one scan the harness runs
Reward	None is computed

The tax agent

Documents, a form, a mail account
What a tool returns
Read, extract, fill, send
None is computed

The harness

The program around the model. The model only writes text; the harness turns that text into actions and feeds back what happened.

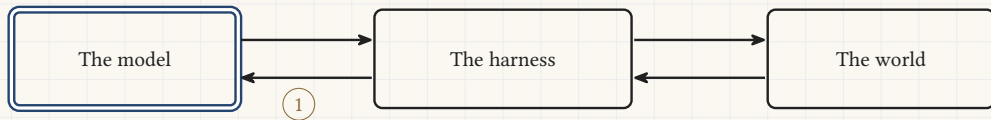


*Anthropic, Claude API documentation,
tool use, 2026*

*Young, Effective harnesses for
long-running agents, Anthropic, 2025*

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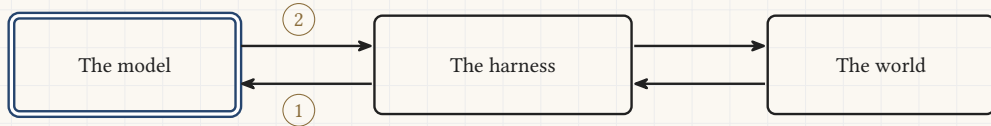
- ① Builds the prompt: the tools and the transcript

*Anthropic, Claude API documentation,
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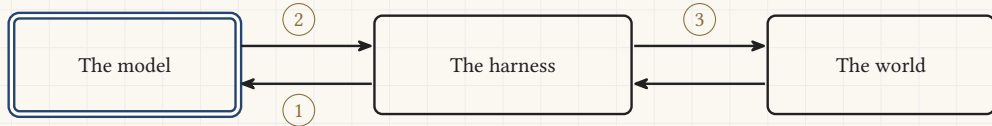
- ① Builds the prompt: the tools and the transcript
- ② Reads the reply and finds the tool call

*Anthropic, Claude API documentation,
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The program around the model. The model only writes text; the harness turns that text into actions and feeds back what happened.



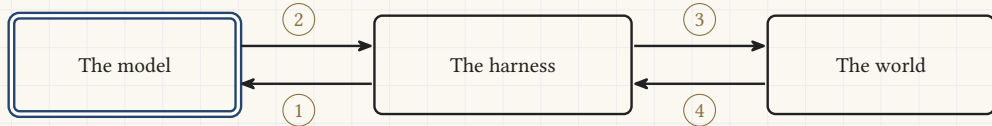
- ① Builds the prompt: the tools and the transcript
- ② Reads the reply and finds the tool call
- ③ Runs that tool on the world

*Anthropic, Claude API documentation,
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The harness

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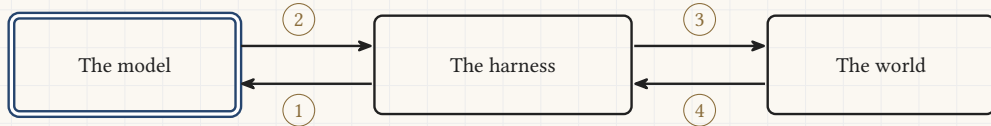
- ① Builds the prompt: the tools and the transcript
- ② Reads the reply and finds the tool call
- ③ Runs that tool on the world
- ④ Appends the result, then calls the model again or stops

*Anthropic, Claude API documentation,
tool use, 2026*

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The harness

The program around the model. The model only writes text; the harness turns that text into actions and feeds back what happened.



In the spy game: `run_agent()`

- ① Builds the prompt: the tools and the transcript
- ② Reads the reply and finds the tool call
- ③ Runs that tool on the world
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*Anthropic, Claude API documentation,
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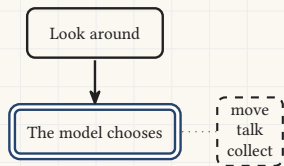
*Young, Effective harnesses for
long-running agents, Anthropic, 2025*

2

Section 2

ReAct, played by hand

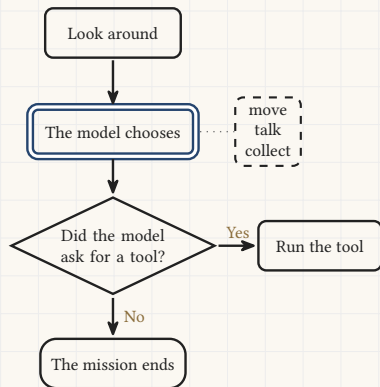
Thought, action, observation



-  A call to the model
-  Something that is not a call to the model
-  The tools the model may choose from

Yao and others, ReAct, ICLR 2023

Thought, action, observation



A call to the model



Something that is not a call to the model



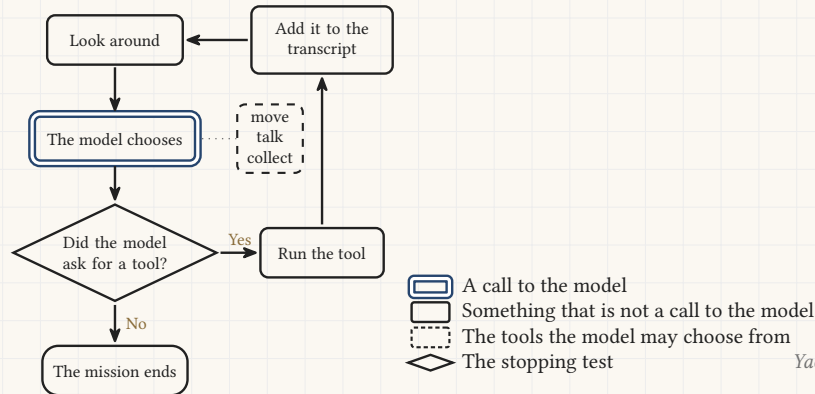
The tools the model may choose from



The stopping test

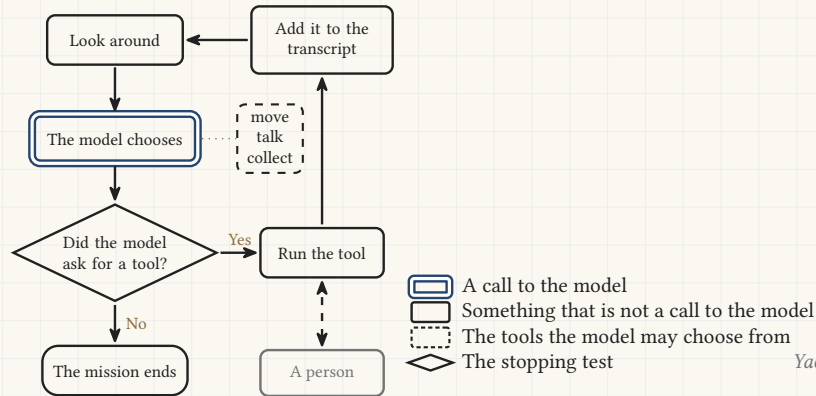
Yao and others, ReAct, ICLR 2023

Thought, action, observation



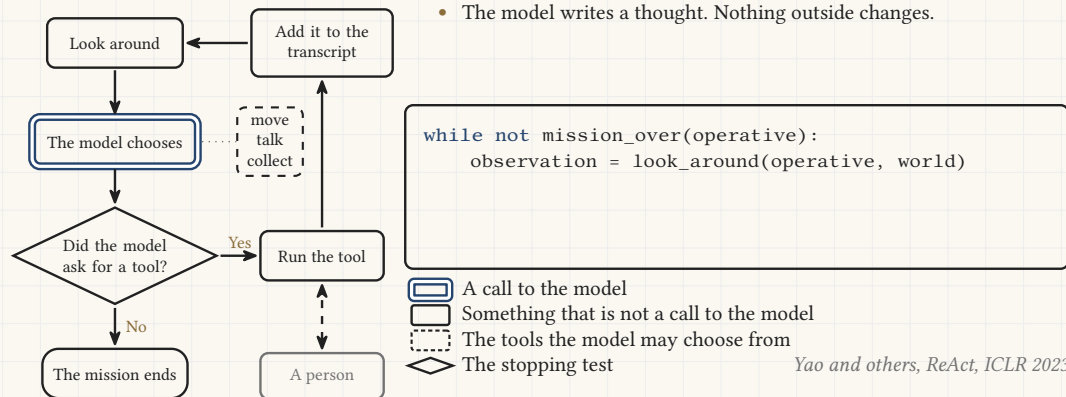
Yao and others, ReAct, ICLR 2023

Thought, action, observation



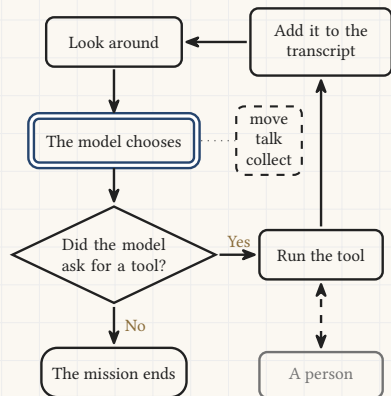
Yao and others, ReAct, ICLR 2023

Thought, action, observation



Yao and others, ReAct, ICLR 2023

Thought, action, observation



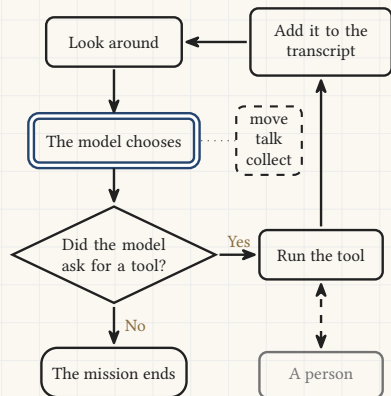
- The model writes a thought. Nothing outside changes.
- It writes one action. The harness runs it.
- What comes back is appended, and the model is called again.

```
while not mission_over(operative):  
    observation = look_around(operative, world)  
    action = decide(briefing, transcript, observation)  
    result = execute(action, tools)  
    remember(transcript, observation, action, result)
```

- A call to the model
- Something that is not a call to the model
- The tools the model may choose from
- The stopping test

Yao and others, ReAct, ICLR 2023

Thought, action, observation



- The model writes a thought. Nothing outside changes.
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while not mission_over(operative):  
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```

- A call to the model
- Something that is not a call to the model
- The tools the model may choose from
- The stopping test

The stopping test is in the loop's own condition. It is not a step counter. *Yao and others, ReAct, ICLR 2023*

The transcript the model is called with



Written once, by a person

You are a spy agent on a 3x3 grid.
Collect 3 dossiers in 30 turns.
Tools (use exactly one per turn):
 TOOL: move(direction="north")
 TOOL: talk(message="hello")
 TOOL: collect()
Respond with exactly one TOOL: line.
Write one THOUGHT: line, then
one TOOL: line.

Yao and others, ReAct, ICLR 2023
Rajasekaran and others, Effective
context engineering, Anthropic, 2025

The transcript the model is called with

The transcript



Written once, by a person

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Yao and others, ReAct, ICLR 2023
Rajasekaran and others, Effective
context engineering, Anthropic, 2025

The transcript the model is called with

Observation

The transcript

The harness pasted the briefing here

[the mission briefing, 23 lines]

You are at position (2, 0).

East: Informant *[the scan's other 4 lines]*

Written once, by a person

You are a spy agent on a 3x3 grid.

Collect 3 dossiers in 30 turns.

Tools (use exactly one per turn):

TOOL: move(direction="north")

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Yao and others, ReAct, ICLR 2023
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The transcript the model is called with

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The transcript

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TOOL: move(direction="north")

TOOL: talk(message="hello")

TOOL: collect()

Respond with exactly one TOOL: line.

Write one THOUGHT: line, then

one TOOL: line.

What the model wrote this step

[no thought on this turn]

Yao and others, ReAct, ICLR 2023
Rajasekaran and others, Effective
context engineering, Anthropic, 2025

The transcript the model is called with

Observation

The transcript

[the mission briefing, 23 lines]

You are at position (2, 0).

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TOOL: collect()

Respond with exactly one TOOL: line.

Write one THOUGHT: line, then

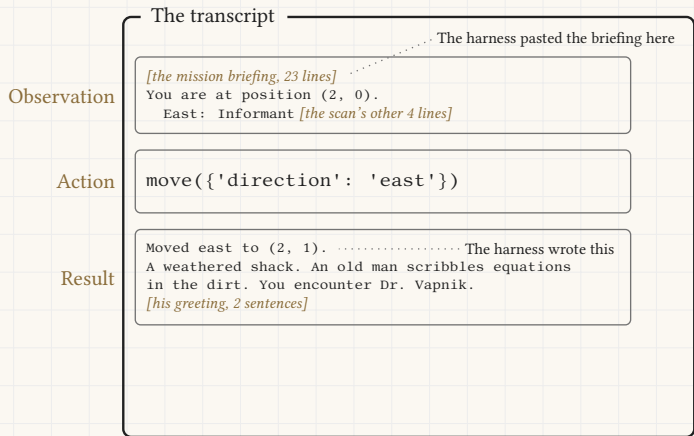
one TOOL: line.

What the model wrote this step

TOOL: move(direction="east")

Yao and others, ReAct, ICLR 2023
Rajasekaran and others, Effective
context engineering, Anthropic, 2025

The transcript the model is called with



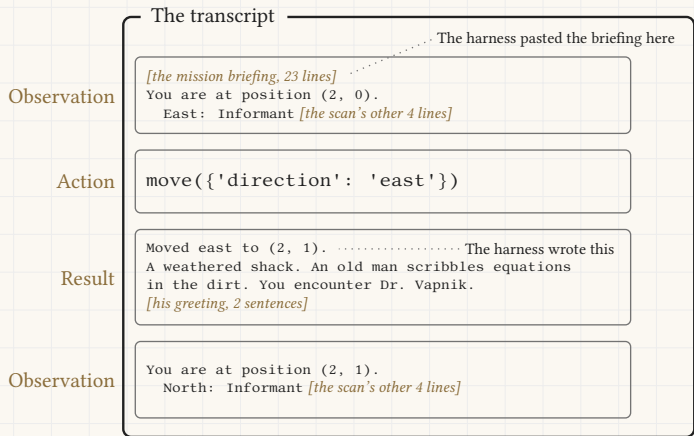
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What the model wrote this step

Yao and others, ReAct, ICLR 2023
Rajasekaran and others, Effective
context engineering, Anthropic, 2025

The transcript the model is called with



Written once, by a person

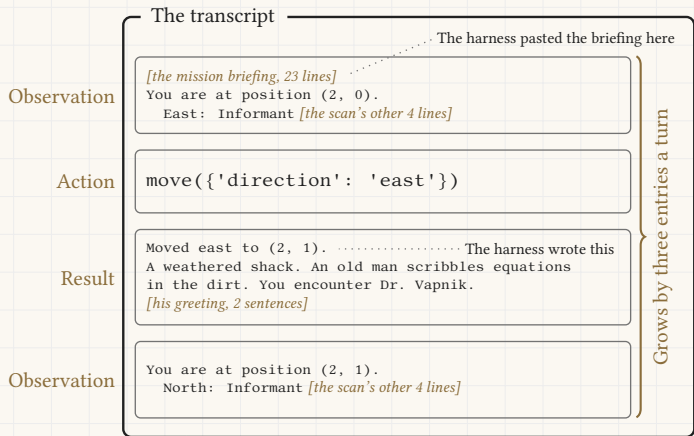
You are a spy agent on a 3x3 grid.
Collect 3 dossiers in 30 turns.
Tools (use exactly one per turn):
TOOL: move(direction="north")
TOOL: talk(message="hello")
TOOL: collect()
Respond with exactly one TOOL: line.
Write one THOUGHT: line, then
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What the model wrote this step

TOOL: talk(message="Dr. Vapnik, I
seek the global minimum. Do you have
any jobs for an agent like me?")

Yao and others, ReAct, ICLR 2023
Rajasekaran and others, Effective
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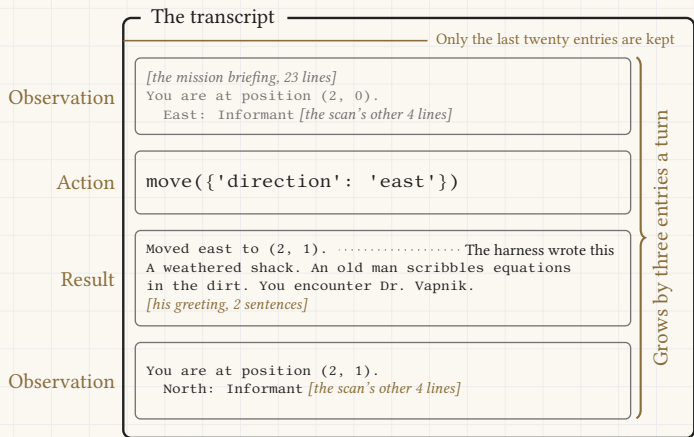
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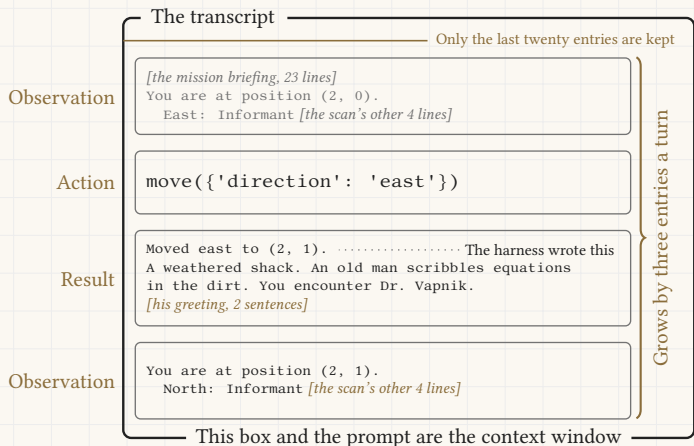
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gemini-2.5-flash, 16 September 2026
Yao and others, ReAct, ICLR 2023
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context engineering, Anthropic, 2025

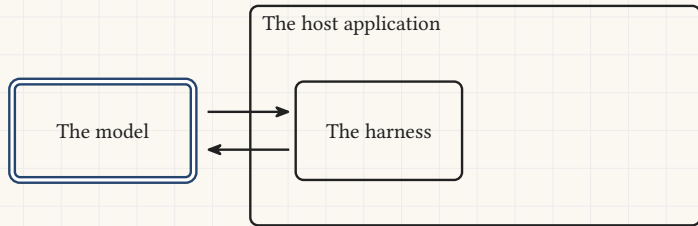
3

Section 3

The Model Context Protocol

What MCP is

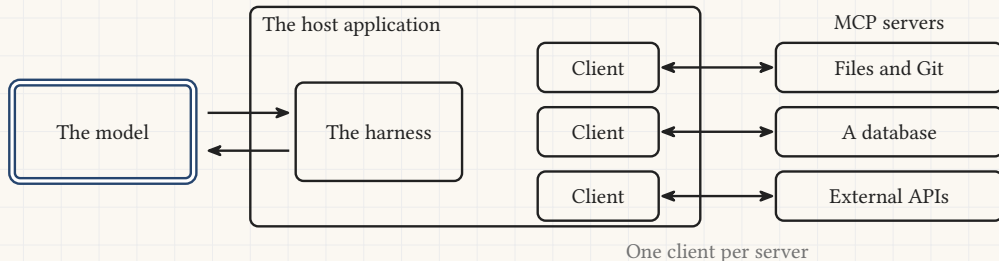
An open protocol: a standard plug between an application that uses a model and the servers that offer it tools and data.



Model Context Protocol, Specification, 2026

What MCP is

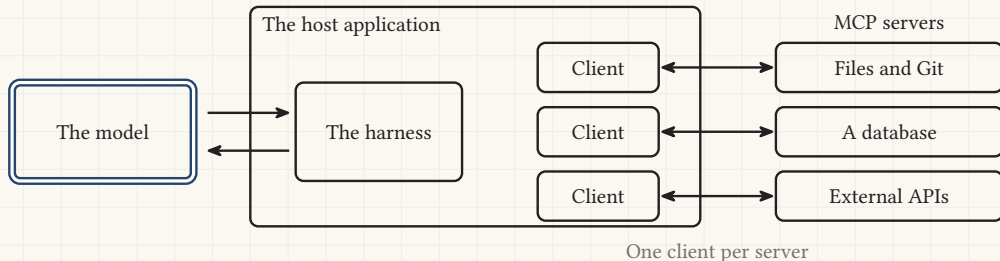
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Model Context Protocol, Specification, 2026

What MCP is

An open protocol: a standard plug between an application that uses a model and the servers that offer it tools and data.



Tools: functions the model may ask for, run by the server

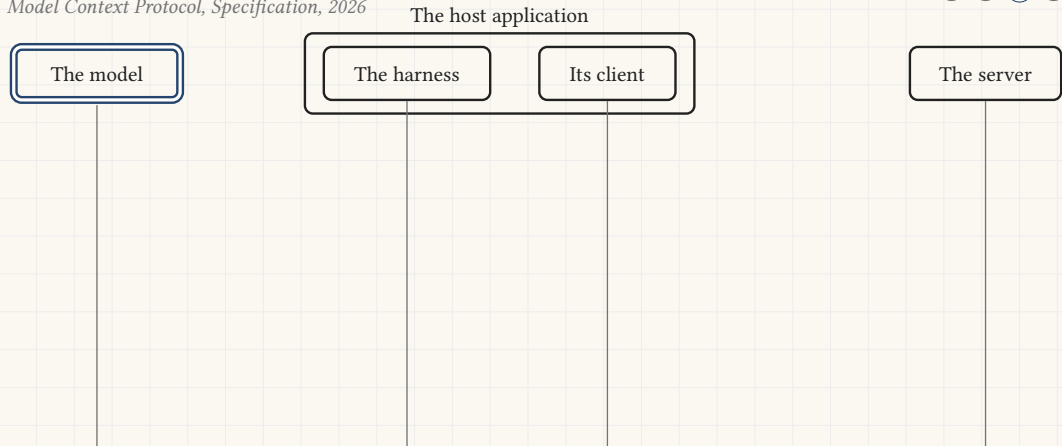
Resources: context and data, for the user or the model

Prompts: templated messages and workflows for the user

Model Context Protocol, Specification, 2026

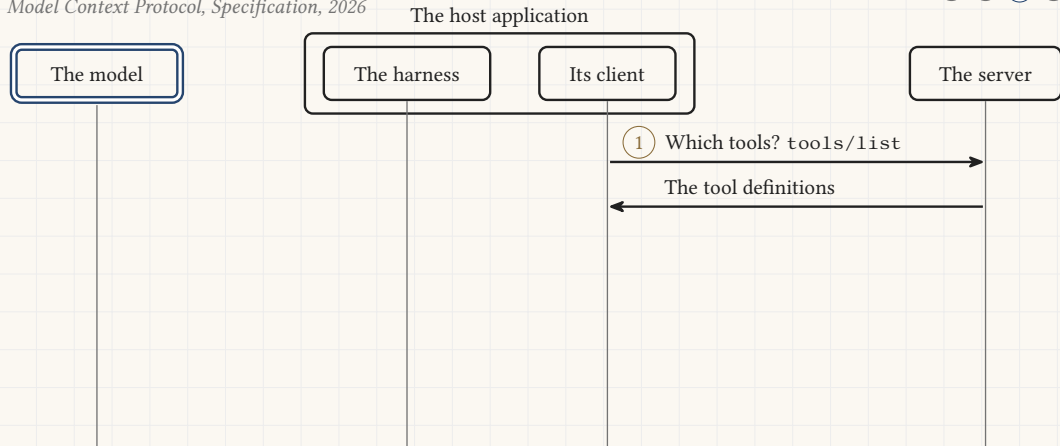
How a tool call travels through MCP

Model Context Protocol, Specification, 2026



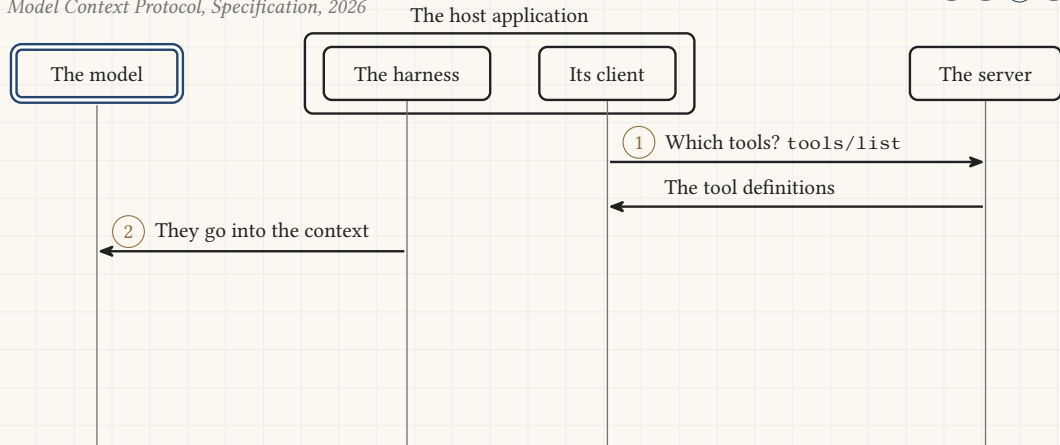
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Model Context Protocol, Specification, 2026



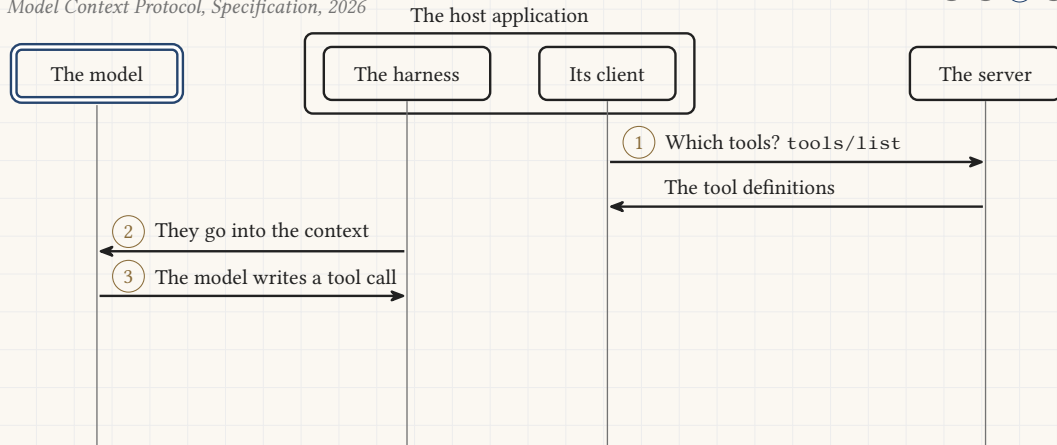
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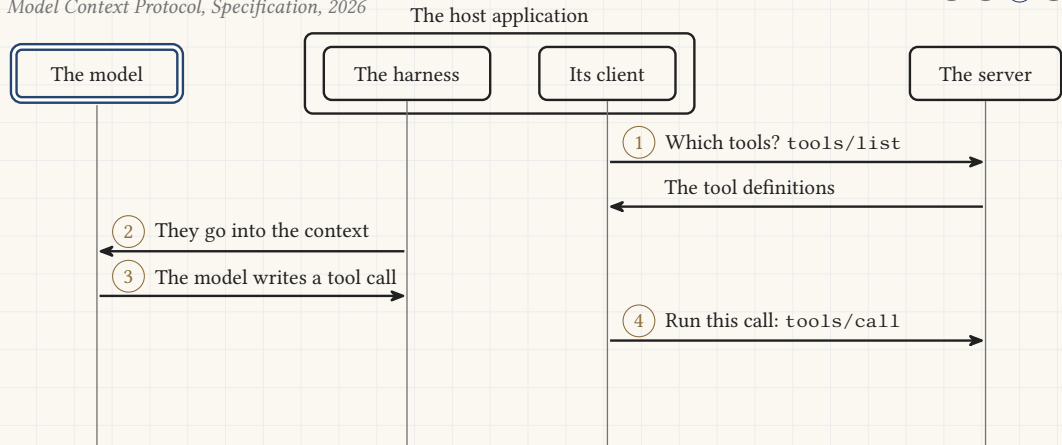
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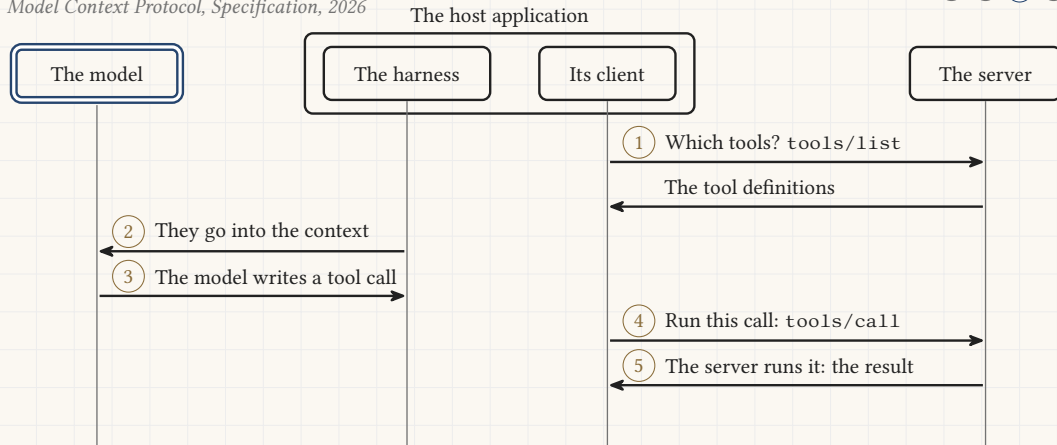
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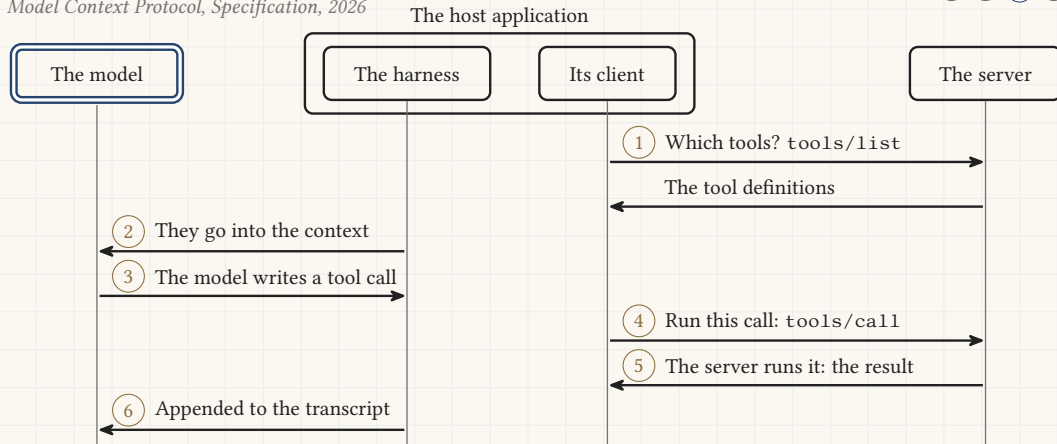
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Model Context Protocol, Specification, 2026



How a tool call travels through MCP

Model Context Protocol, Specification, 2026



4

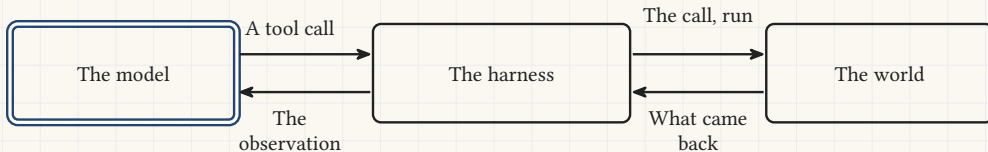
Section 4

The harness

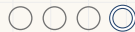
An agent is a system



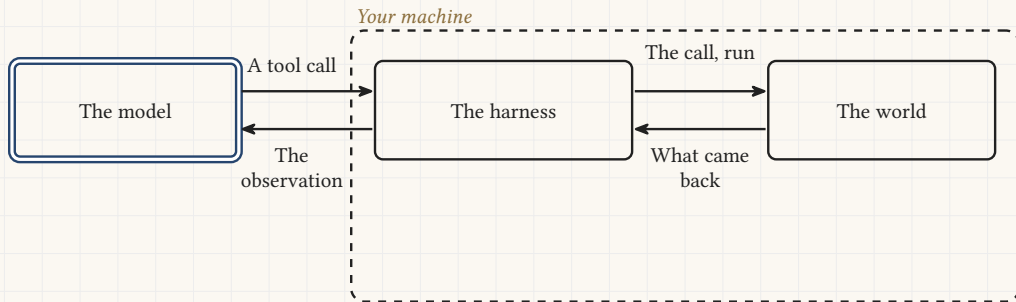
Cherny, Claude Code best practices, Anthropic, 2025
Anthropic, Claude API documentation, tool use, 2026



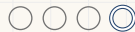
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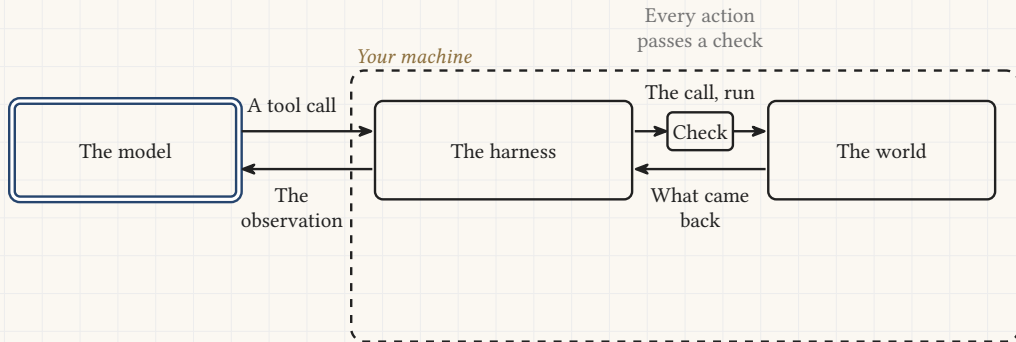
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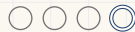
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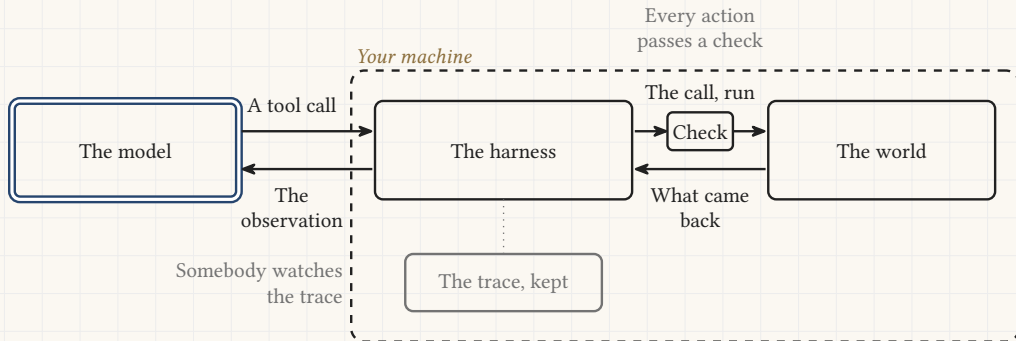
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Play the spy game



[https://eth-bmai-hs26.github.io/
BMAI-PAGE/viz/we2/spy-game/](https://eth-bmai-hs26.github.io/BMAI-PAGE/viz/we2/spy-game/)

- 1 Open the page in your browser
- 2 Paste your OpenRouter key and press Save
- 3 Play with Manual, or choose Auto and press Run Agent

References: definitions and the loop

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